STAT429-Project-240504

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Preprocessing

```r
library(readxl)
library(astsa)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

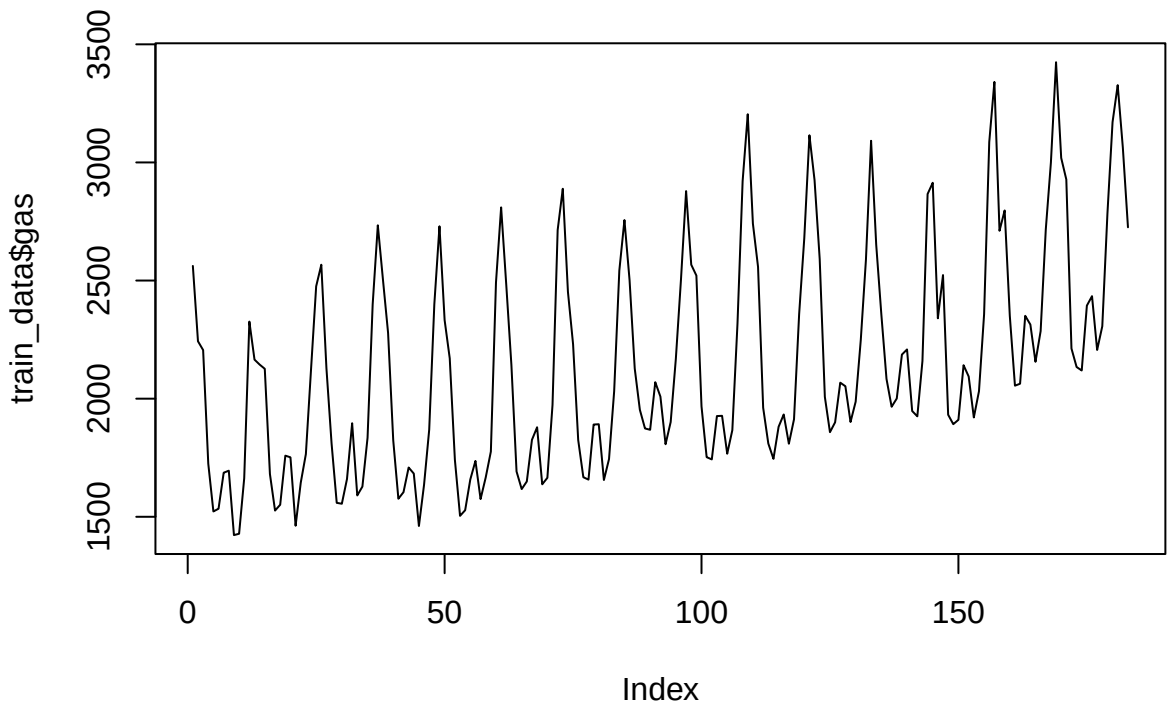
```r
data <- read_excel("/Users/ricky/Desktop/STAT 429/Project/from-2005-data.xlsx")
colnames(data) <- c("date", "gas", "import", "coal", "elec")
data$date <- as.Date(data$date)
data$time_trend <- 1:nrow(data)

# Split into Training (first 80%) and Testing set (last 20%).
training_size <- floor(0.8 * nrow(data))
train_data <- data[1:training_size, ]
test_data <- data[(training_size + 1):nrow(data), ]
head(train_data)
```

```
## # A tibble: 6 x 6
##   date         gas import   coal  elec time_trend
##   <date>     <dbl>  <dbl>  <dbl> <dbl>      <int>
## 1 2005-01-15 2562.   405. 99403.  343.          1
## 2 2005-02-15 2243.   356. 87721.  298.          2
## 3 2005-03-15 2206.   380. 91151.  317.          3
## 4 2005-04-15 1725.   326. 80860.  290.          4
## 5 2005-05-15 1523.   334. 86368.  315.          5
## 6 2005-06-15 1534.   322. 96504.  364.          6
```

```r
head(test_data)
```

```
## # A tibble: 6 x 6
##   date         gas import   coal  elec time_trend
##   <date>     <dbl>  <dbl>  <dbl> <dbl>      <int>
## 1 2020-04-15 2264.   190. 26754.  280.        184
## 2 2020-05-15 2077.   187. 29784.  305.        185
## 3 2020-06-15 2134.   187. 39798.  352.        186
## 4 2020-07-15 2468.   210. 52852.  410.        187
## 5 2020-08-15 2404.   211. 53610.  399.        188
## 6 2020-09-15 2157.   174. 41828.  333.        189
```

```r
plot(train_data$gas, type = "l") # Linear trend or quadratic trend
```




```r
# Main regression models
data$time_trend <- 1:nrow(data)
## Regression 1
lm_model_1 <- lm(gas ~ time_trend + import + coal + elec, data = train_data) # Linear trend
summary(lm_model_1)
```

```
##
## Call:
## lm(formula = gas ~ time_trend + import + coal + elec, data = train_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -677.75 -213.08  -36.31  189.74  743.82
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.021e+02  3.321e+02  -2.716  0.00725 **
## time_trend   1.776e+01  1.519e+00  11.695  < 2e-16 ***
## import       7.661e+00  7.120e-01  10.761  < 2e-16 ***
## coal         2.345e-02  4.554e-03   5.149 6.88e-07 ***
## elec        -7.476e+00  1.241e+00  -6.025 9.46e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 303.8 on 178 degrees of freedom
## Multiple R-squared:  0.5752, Adjusted R-squared:  0.5657
## F-statistic: 60.27 on 4 and 178 DF,  p-value: < 2.2e-16
```

```r
# Regression 2
lm_model_2 <- lm(gas ~ I(time_trend^2) + import + coal + elec, data = train_data) # Quadratic trend
```

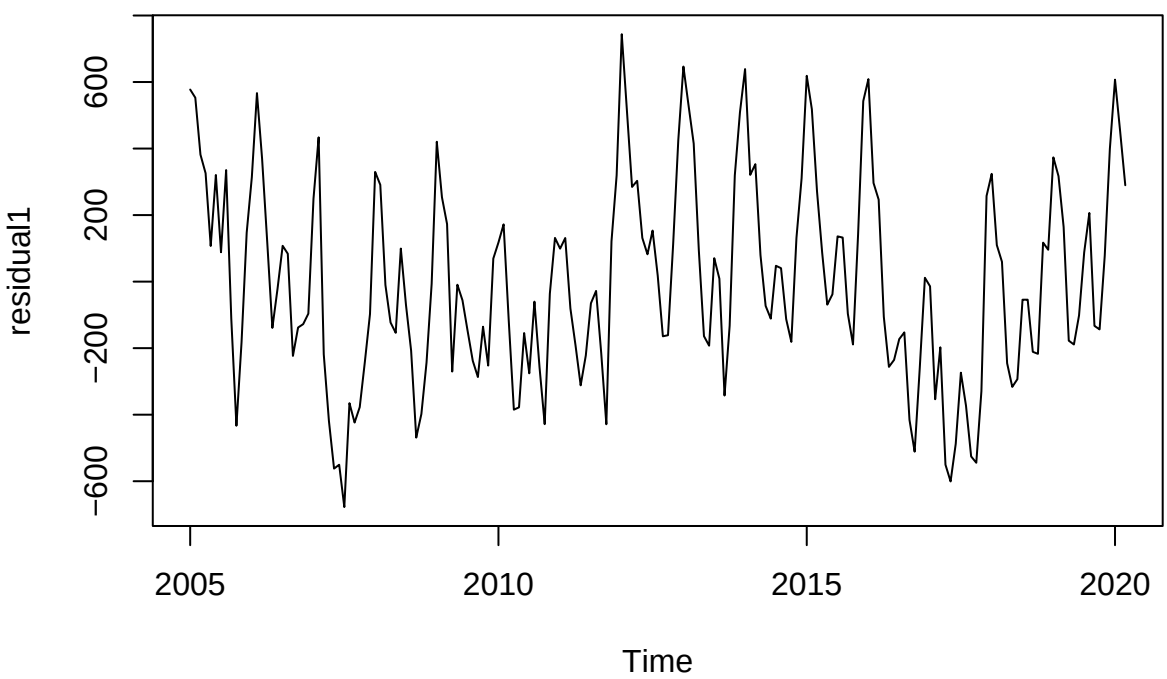
```r
summary(lm_model_2)
```

```
##
## Call:
## lm(formula = gas ~ I(time_trend^2) + import + coal + elec, data = train_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -723.95 -226.74  -78.08  200.15  924.79
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)      687.744555 311.272240   2.209   0.0284 *
## I(time_trend^2)    0.078449   0.009501   8.257 3.27e-14 ***
## import             3.644404   0.709137   5.139 7.21e-07 ***
## coal               0.026446   0.006477   4.083 6.70e-05 ***
## elec              -7.261139   1.605886  -4.522 1.12e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 343.6 on 178 degrees of freedom
## Multiple R-squared:  0.4569, Adjusted R-squared:  0.4447
## F-statistic: 37.44 on 4 and 178 DF,  p-value: < 2.2e-16
```

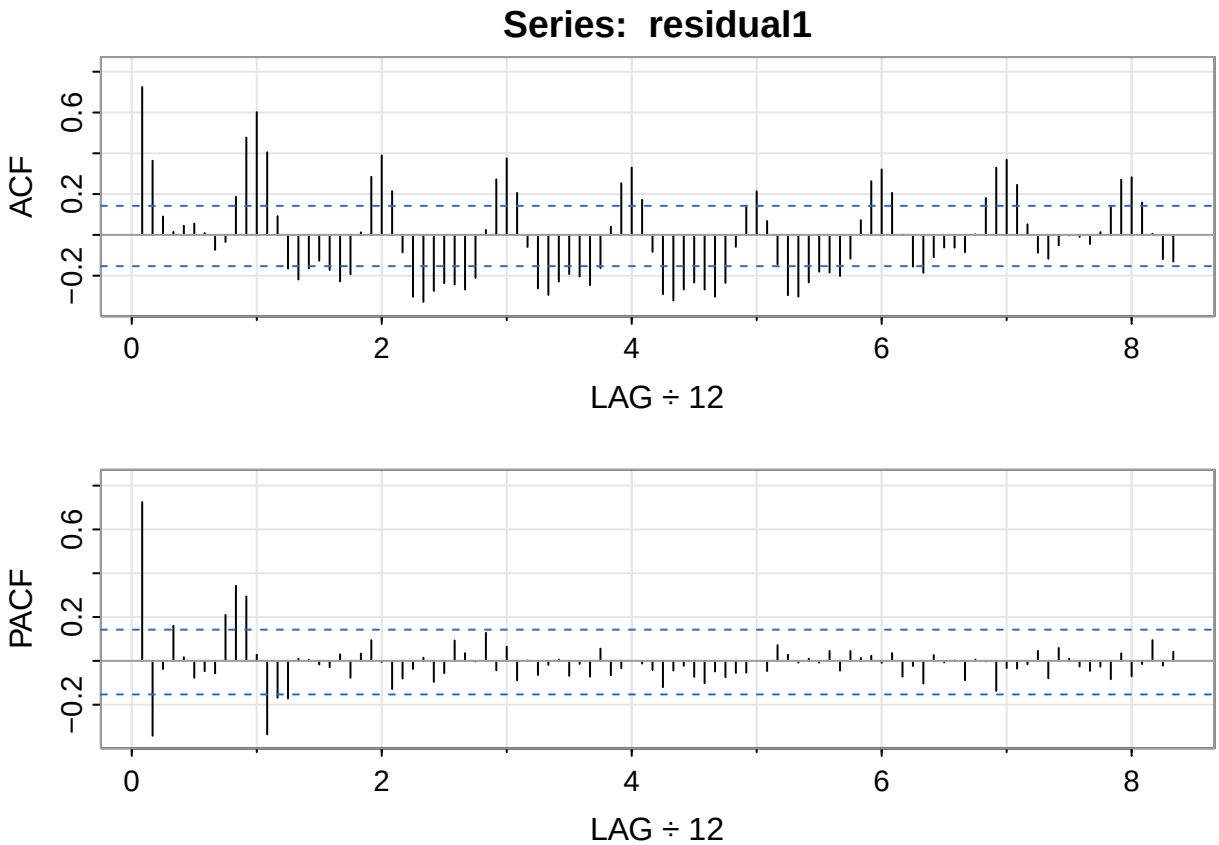
```r
residual1 = ts(residuals(lm_model_1), frequency = 12, start = c(2005, 1))
residual2 = ts(residuals(lm_model_2), frequency = 12, start = c(2005, 1))
```

Transformation for residual1

```r
# Plotting & ACF & PACF
ts.plot(residual1)
```



```r
acf2(residual1, max.lag = 100) # clear seasonality, with s = 12, clear non-stationarity
```




```
##      [,1]  [,2]  [,3] [,4] [,5]  [,6]  [,7]  [,8]  [,9] [,10] [,11] [,12] [,13]
## ACF  0.72  0.36  0.09 0.02 0.04  0.06  0.01 -0.07 -0.03  0.19  0.48  0.60  0.41
## PACF 0.72 -0.34 -0.04 0.16 0.02 -0.08 -0.05 -0.06  0.21  0.34  0.29  0.03 -0.34
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF   0.09 -0.16 -0.22 -0.16 -0.13 -0.17 -0.23 -0.19  0.01  0.28  0.39  0.21
## PACF -0.17 -0.17  0.01  0.00 -0.02 -0.03  0.03 -0.08  0.03  0.09  0.00 -0.13
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF  -0.09 -0.30 -0.33 -0.27 -0.24 -0.24 -0.27 -0.21  0.02  0.27  0.38  0.21
## PACF -0.08 -0.04  0.01 -0.10 -0.06  0.09  0.03  0.00  0.13 -0.04  0.06 -0.09
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48] [,49]
## ACF  -0.06 -0.26 -0.29 -0.23 -0.19 -0.20 -0.25 -0.16  0.04  0.25  0.33  0.17
## PACF  0.00 -0.06 -0.02  0.01 -0.07 -0.01 -0.07  0.06 -0.07 -0.03  0.00 -0.01
##      [,50] [,51] [,52] [,53] [,54] [,55] [,56] [,57] [,58] [,59] [,60] [,61]
## ACF  -0.08 -0.29 -0.32 -0.27 -0.23 -0.27 -0.30 -0.24 -0.06  0.14  0.21  0.07
## PACF -0.04 -0.12 -0.04 -0.02 -0.07 -0.10 -0.05 -0.07 -0.05 -0.05  0.00 -0.05
##      [,62] [,63] [,64] [,65] [,66] [,67] [,68] [,69] [,70] [,71] [,72] [,73]
## ACF  -0.14 -0.30 -0.30 -0.23 -0.18 -0.18 -0.20 -0.12  0.07  0.26  0.32  0.21
## PACF  0.07  0.03 -0.01  0.01 -0.01  0.05 -0.04  0.04  0.01  0.02 -0.01  0.04
##      [,74] [,75] [,76] [,77] [,78] [,79] [,80] [,81] [,82] [,83] [,84] [,85]
## ACF   0.00 -0.16 -0.19 -0.11 -0.06 -0.06 -0.08  0.00  0.18  0.33  0.37  0.25
## PACF -0.07 -0.02 -0.10  0.03 -0.01  0.00 -0.09  0.01  0.00 -0.14 -0.03 -0.03
##      [,86] [,87] [,88] [,89] [,90] [,91] [,92] [,93] [,94] [,95] [,96] [,97]
## ACF   0.05 -0.09 -0.12 -0.05  0.00 -0.01 -0.04  0.01  0.14  0.27  0.28  0.16
## PACF -0.02  0.05 -0.08  0.06  0.01 -0.03 -0.05 -0.03 -0.08  0.03 -0.07 -0.01
##      [,98] [,99] [,100]
```

```
## ACF   0.01 -0.12  -0.13
## PACF  0.09 -0.02   0.04
```

```r
# adf test: stationary, but discard!
adf.test(residual1)
```

```
## Warning in adf.test(residual1): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  residual1
## Dickey-Fuller = -4.6743, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

```r
# kpss test: not trend stationary
kpss.test(residual1, null = "Level") # stationary, but discard!
```

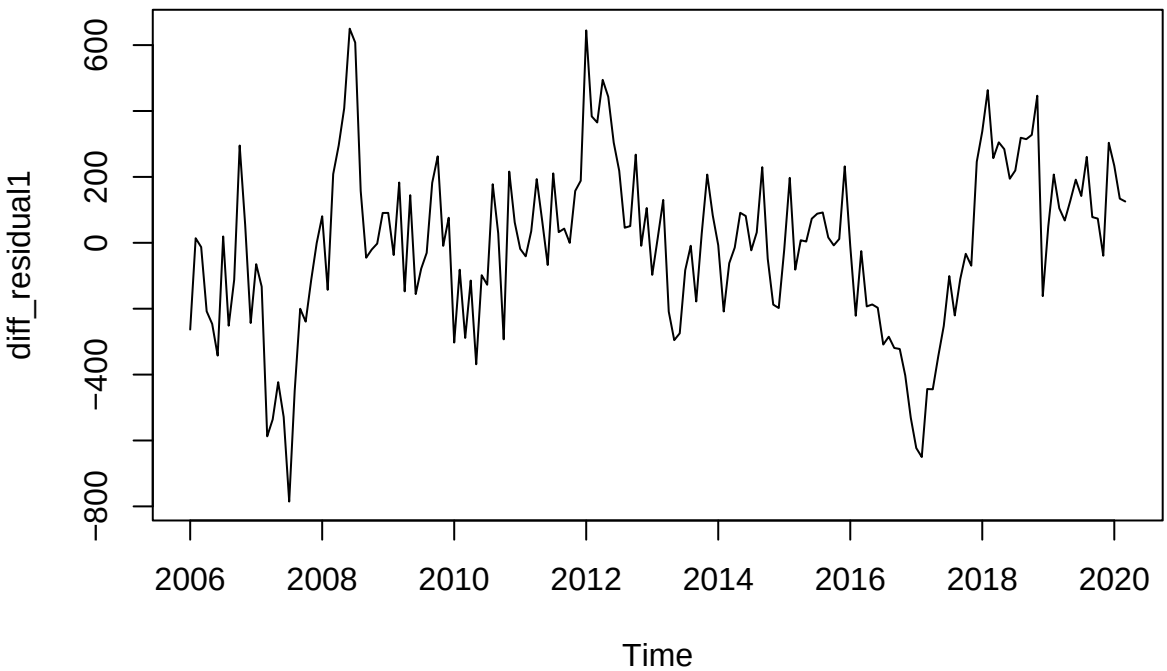
```
## Warning in kpss.test(residual1, null = "Level"): p-value greater than printed
## p-value
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  residual1
## KPSS Level = 0.12116, Truncation lag parameter = 4, p-value = 0.1
```

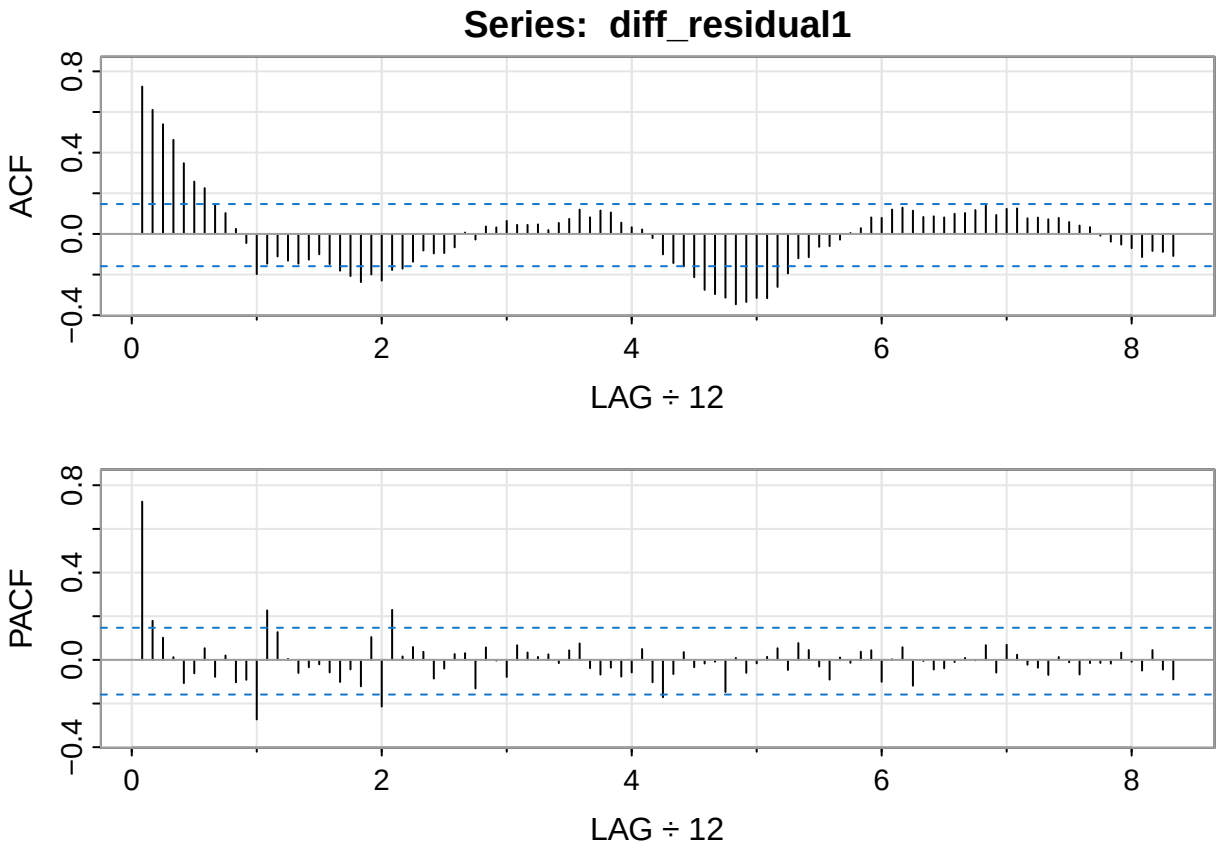
```r
kpss.test(residual1, null = "Trend") # but not trend stationary
```

```
##
##  KPSS Test for Trend Stationarity
##
## data:  residual1
## KPSS Trend = 0.12116, Truncation lag parameter = 4, p-value = 0.09599
```

```r
# Transformation: diff and seasonal diff
diff_residual1 = diff(residual1, 12) # seasonal differencing
ts.plot(diff_residual1)
```



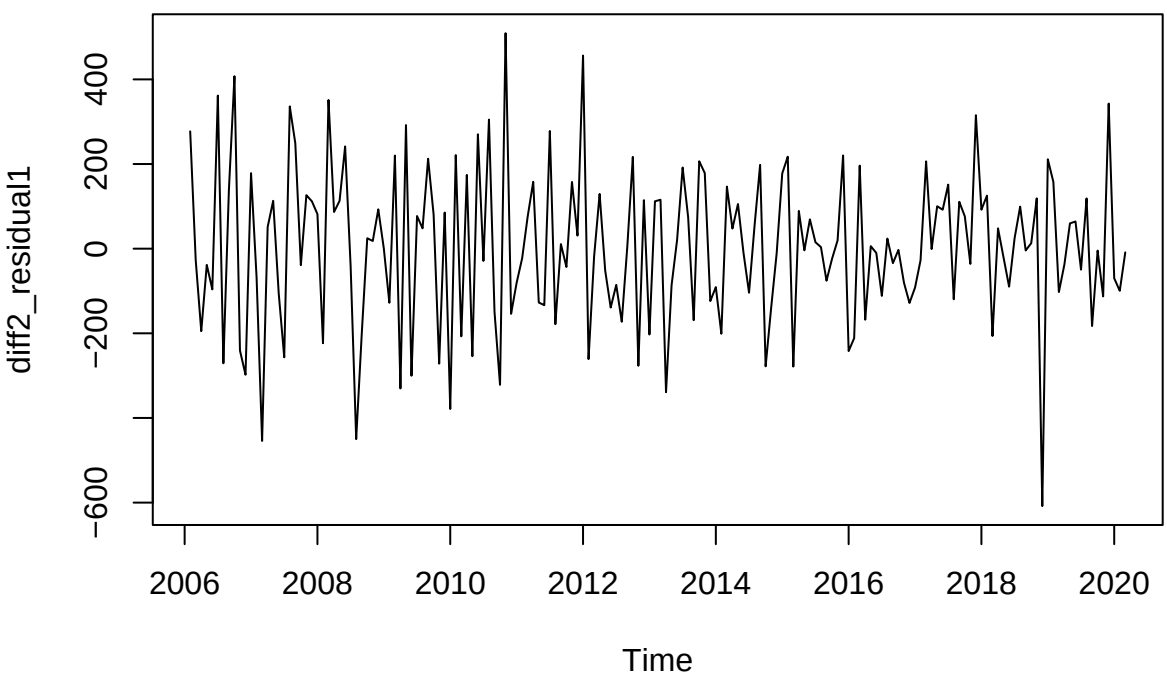

```r
acf2(diff_residual1, max.lag = 100) # ACF still decaying very slowly, another differencing
```



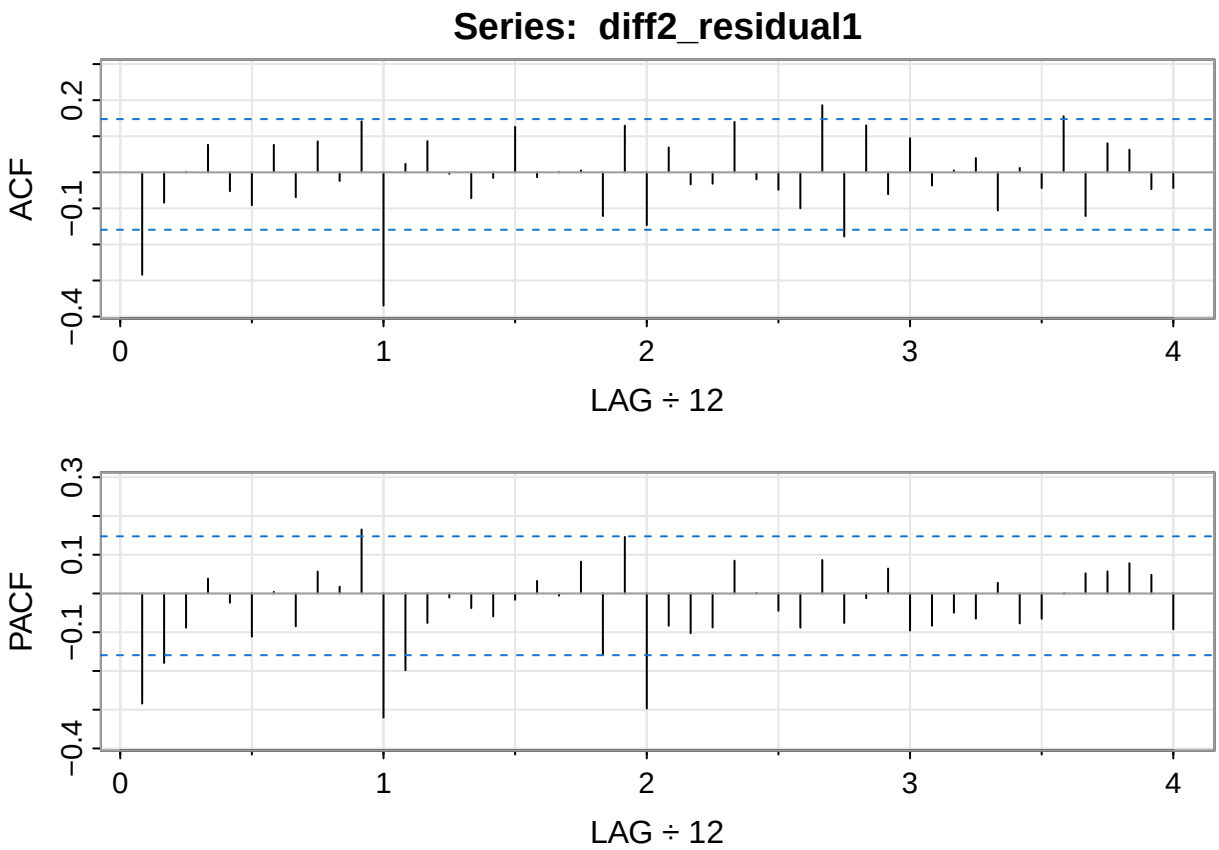

```
##      [,1] [,2] [,3] [,4]  [,5]  [,6] [,7]  [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  0.72 0.61 0.54 0.46  0.35  0.26 0.23  0.14 0.10  0.02 -0.04 -0.20 -0.15
## PACF 0.72 0.18 0.10 0.01 -0.11 -0.06 0.05 -0.08 0.02 -0.10 -0.09 -0.27  0.23
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
```

```
## ACF  -0.11 -0.13 -0.15 -0.13 -0.10 -0.15 -0.18 -0.21 -0.24  -0.2 -0.23 -0.18
## PACF  0.13  0.00 -0.06 -0.03 -0.02 -0.06 -0.10 -0.04 -0.12   0.1 -0.21  0.23
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF  -0.17 -0.14 -0.08 -0.10 -0.09 -0.07  0.01 -0.03  0.04  0.03  0.06  0.04
## PACF  0.02  0.06  0.04 -0.09 -0.04  0.03  0.03 -0.13  0.06  0.00 -0.08  0.07
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48] [,49]
## ACF   0.04  0.05  0.02  0.05  0.07  0.12  0.08  0.12  0.11  0.06  0.03  0.02
## PACF  0.03  0.01  0.03 -0.01  0.04  0.08 -0.04 -0.07 -0.04 -0.08 -0.06  0.05
##      [,50] [,51] [,52] [,53] [,54] [,55] [,56] [,57] [,58] [,59] [,60] [,61]
## ACF  -0.02 -0.10 -0.14 -0.16 -0.21 -0.28 -0.30 -0.31 -0.35 -0.33 -0.32 -0.32
## PACF -0.10 -0.17 -0.06  0.04 -0.03 -0.02 -0.01 -0.15  0.01 -0.06 -0.02  0.01
##      [,62] [,63] [,64] [,65] [,66] [,67] [,68] [,69] [,70] [,71] [,72] [,73]
## ACF  -0.26 -0.19 -0.12 -0.11 -0.06 -0.06 -0.03  0.00  0.03  0.08  0.08  0.12
## PACF  0.05 -0.05  0.08  0.05 -0.03 -0.09  0.01 -0.01  0.04  0.04 -0.10  0.00
##      [,74] [,75] [,76] [,77] [,78] [,79] [,80] [,81] [,82] [,83] [,84] [,85]
## ACF   0.13  0.11  0.08  0.09  0.08  0.10  0.10  0.12  0.14  0.09  0.12  0.13
## PACF  0.06 -0.12 -0.01 -0.04 -0.04 -0.01  0.01  0.00  0.07 -0.06  0.07  0.02
##      [,86] [,87] [,88] [,89] [,90] [,91] [,92] [,93] [,94] [,95] [,96] [,97]
## ACF   0.08  0.08  0.07  0.08  0.06  0.04  0.03 -0.01 -0.04 -0.05 -0.07 -0.11
## PACF -0.02 -0.04 -0.07  0.01 -0.01 -0.07 -0.01 -0.01 -0.02  0.03 -0.01 -0.05
##      [,98] [,99] [,100]
## ACF  -0.08 -0.09  -0.11
## PACF  0.04 -0.04  -0.09
```

```r
diff2_residual1 = diff(diff(residual1, 12))
ts.plot(diff2_residual1)
```




```r
acf2(diff2_residual1)
```




```
##       [,1]  [,2]  [,3] [,4]  [,5]  [,6] [,7]  [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  -0.28 -0.08  0.00 0.08 -0.05 -0.09 0.08 -0.07 0.09 -0.02  0.14 -0.37  0.02
## PACF -0.28 -0.18 -0.09 0.04 -0.02 -0.11 0.00 -0.08 0.06  0.02  0.17 -0.32 -0.20
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF   0.09  0.00 -0.07 -0.02  0.13 -0.01  0.00  0.01 -0.12  0.13 -0.15  0.07
## PACF -0.08 -0.01 -0.04 -0.06 -0.02  0.03 -0.01  0.08 -0.16  0.15 -0.30 -0.08
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF  -0.03 -0.03  0.14 -0.02 -0.05 -0.10  0.19 -0.18  0.13 -0.06  0.09 -0.04
## PACF -0.10 -0.09  0.08  0.00 -0.04 -0.09  0.09 -0.08 -0.01  0.06 -0.10 -0.08
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48]
## ACF   0.01  0.04 -0.11  0.01 -0.04  0.16 -0.12  0.08  0.06 -0.05 -0.04
## PACF -0.05 -0.07  0.03 -0.08 -0.07  0.00  0.05  0.06  0.08  0.05 -0.09
```

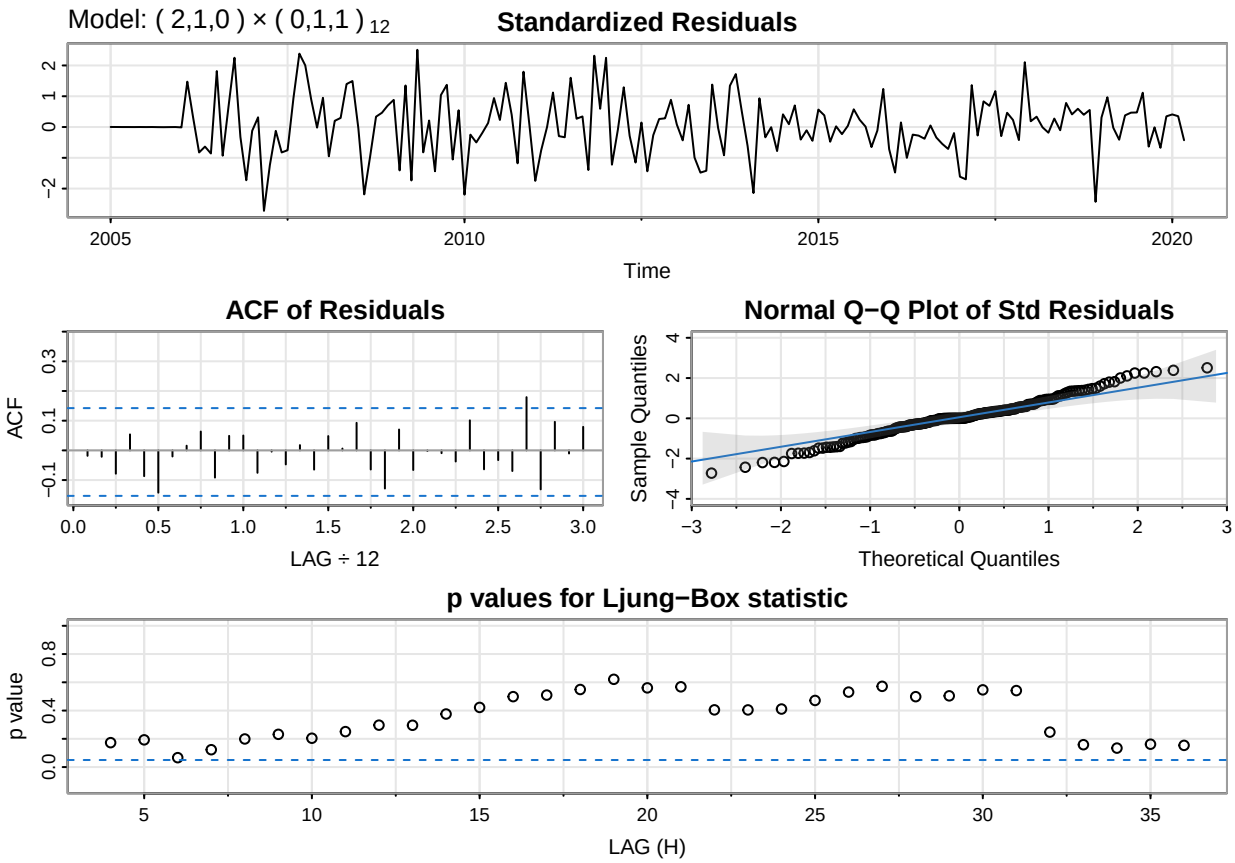
Comments:

- We split the orginal data into a training and a testing set. The training set accounts for the first 80% of the series and the testing set accounts for the last 20% of the series.
- We run 2 major regression on the independent variables and different time trends: linear and quadratic. Then, we obtain the residuals of each regression.
- We first plot acf and pacf for the residuals from regression 1. We see that there seems to be a clear seasonality in the data, with the significant seasonal lags spiking at a frequency `s` of 12. Also, the KPSS test showed that the data is clearly not trend stationary and **ACF is decaying very slowly at seasonal peak** here. Seasonal differencing is required here.
- After the first differencing, we plot and see there are still some pattern in the residuals. Meanwhile, **the ACF of the differenced series is still decaying very slow**, we try an extra normal differencing.
- After that, the data becomes stationary. Both ADF and KPSS tests are passed. Additionally, pacf and acf are not showing any obvious evidence for non-stationarity and seasonality. The preprocessing is done.

Modeling for autocorrelation in residual1

```r
## First, try the interpretability of the acf pacf plots
mod1 = sarima(residual1, p=2, d=1, q=0, P=0, D=1, Q=1, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.218889
## iter   2 value 5.052011
## iter   3 value 5.030874
## iter   4 value 5.024209
## iter   5 value 5.015726
## iter   6 value 5.014220
## iter   7 value 5.014054
## iter   8 value 5.014048
## iter   8 value 5.014048
## final  value 5.014048
## converged
## initial  value 4.991335
## iter   2 value 4.981934
## iter   3 value 4.981850
## iter   4 value 4.981844
## iter   4 value 4.981844
## iter   4 value 4.981844
## final  value 4.981844
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE  t.value p.value
## ar1   -0.3492 0.0748  -4.6662  0.0000
## ar2   -0.2129 0.0746  -2.8527  0.0049
## sma1  -0.7360 0.0647 -11.3724  0.0000
##
## sigma^2 estimated as 20081.85 on 167 degrees of freedom
##
## AIC = 12.84862  AICc = 12.84947  BIC = 12.92241
##
```



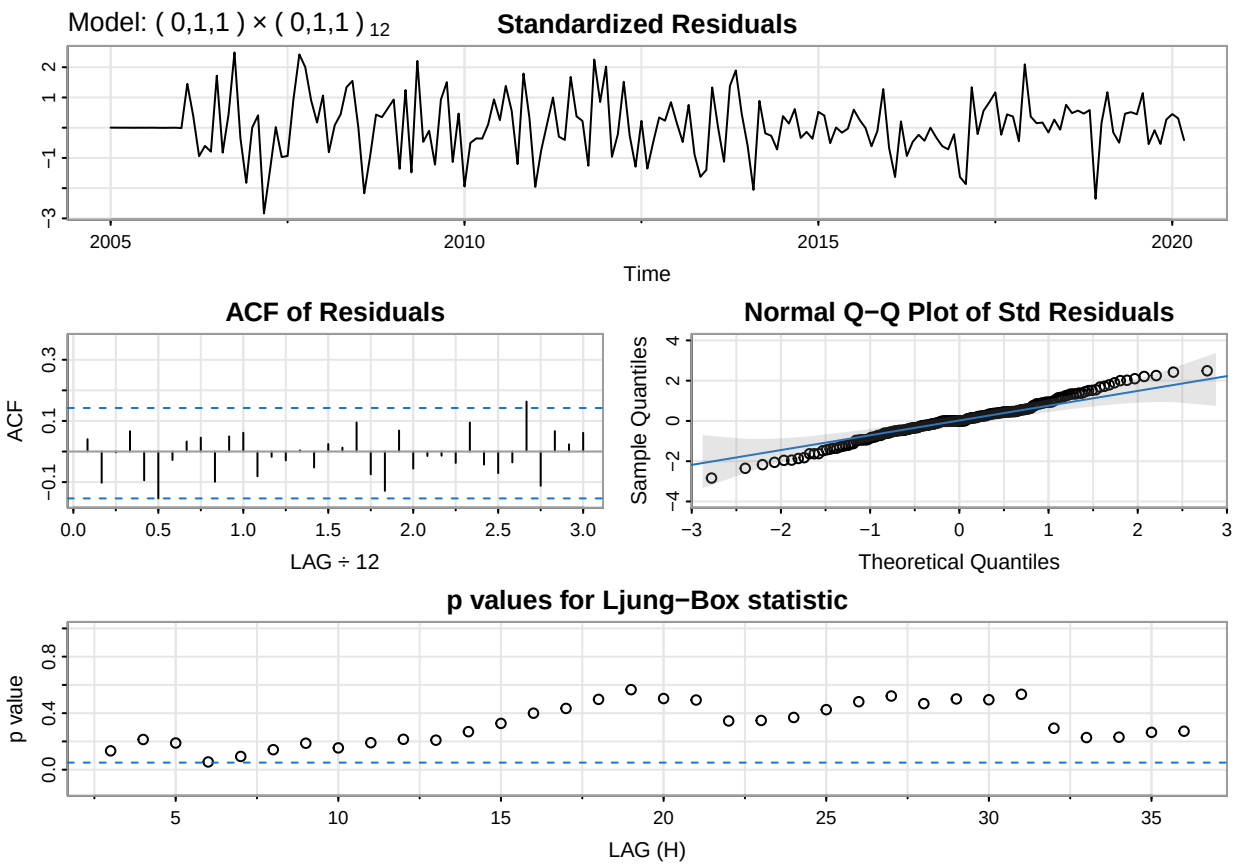

```r
mod1$ttable # all terms are significant
```

```
##      Estimate     SE  t.value p.value
## ar1   -0.3492 0.0748  -4.6662  0.0000
## ar2   -0.2129 0.0746  -2.8527  0.0049
## sma1  -0.7360 0.0647 -11.3724  0.0000
```

```r
mod2 = sarima(residual1, p=0, d=1, q=1, P=0, D=1, Q=1, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.219689
## iter   2 value 5.053338
## iter   3 value 5.030259
## iter   4 value 5.024991
## iter   5 value 5.012906
## iter   6 value 5.010208
## iter   7 value 5.009742
## iter   8 value 5.009664
## iter   9 value 5.009663
## iter   9 value 5.009663
## iter   9 value 5.009663
## final  value 5.009663
## converged
## initial  value 4.989291
## iter   2 value 4.983992
## iter   3 value 4.983906
## iter   4 value 4.983903
## iter   4 value 4.983903
```

```
## iter   4 value 4.983903
## final  value 4.983903
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE  t.value p.value
## ma1   -0.4118 0.0772  -5.3365       0
## sma1  -0.7423 0.0658 -11.2798       0
##
## sigma^2 estimated as 20134.78 on 168 degrees of freedom
##
## AIC = 12.84098  AICc = 12.8414  BIC = 12.89631
##
```



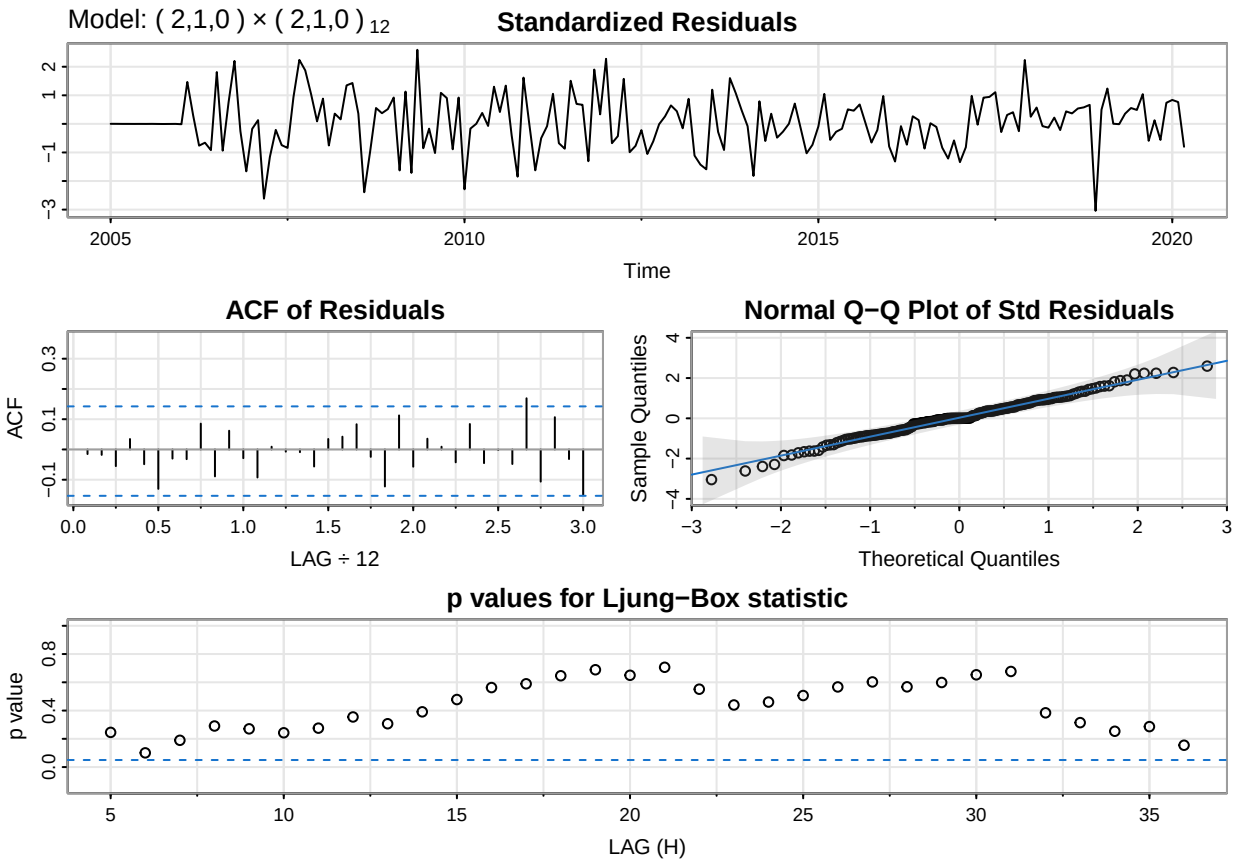

```r
mod2$ttable # all terms are significant
```

```
##      Estimate     SE  t.value p.value
## ma1   -0.4118 0.0772  -5.3365       0
## sma1  -0.7423 0.0658 -11.2798       0
```

```r
mod3 = sarima(residual1, p=2, d=1, q=0, P=2, D=1, Q=0, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.166766
## iter   2 value 4.971743
## iter   3 value 4.930326
## iter   4 value 4.924673
## iter   5 value 4.924659
```

```
## iter   6 value 4.924575
## iter   7 value 4.924574
## iter   7 value 4.924574
## iter   7 value 4.924574
## final  value 4.924574
## converged
## initial  value 4.996773
## iter   2 value 4.993647
## iter   3 value 4.993204
## iter   4 value 4.993180
## iter   5 value 4.993169
## iter   5 value 4.993169
## iter   5 value 4.993169
## final  value 4.993169
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE t.value p.value
## ar1   -0.3595 0.0748 -4.8083   0e+00
## ar2   -0.2579 0.0752 -3.4312   8e-04
## sar1  -0.5885 0.0730 -8.0562   0e+00
## sar2  -0.4226 0.0750 -5.6311   0e+00
##
## sigma^2 estimated as 20826.94 on 166 degrees of freedom
##
## AIC = 12.88304  AICc = 12.88446  BIC = 12.97527
##
```



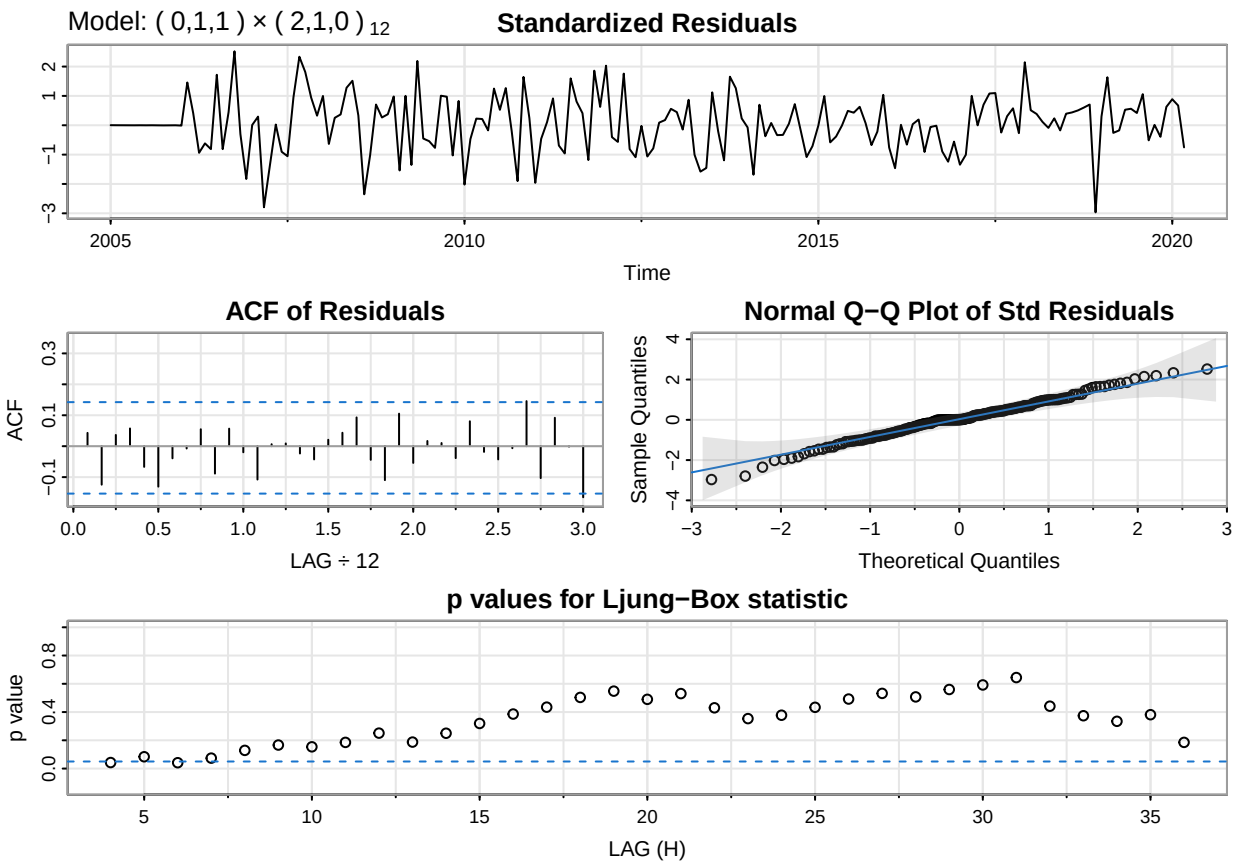

```r
mod3$ttable # all terms are significant
```

```
##      Estimate     SE t.value p.value
## ar1   -0.3595 0.0748 -4.8083   0e+00
## ar2   -0.2579 0.0752 -3.4312   8e-04
## sar1  -0.5885 0.0730 -8.0562   0e+00
## sar2  -0.4226 0.0750 -5.6311   0e+00
```

```r
mod4 = sarima(residual1, p=0, d=1, q=1, P=2, D=1, Q=0, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.179073
## iter   2 value 4.954894
## iter   3 value 4.935769
## iter   4 value 4.917134
## iter   5 value 4.916458
## iter   6 value 4.916351
## iter   7 value 4.916333
## iter   8 value 4.916333
## iter   8 value 4.916333
## final  value 4.916333
## converged
## initial  value 5.005331
## iter   2 value 5.001320
## iter   3 value 5.001163
## iter   4 value 5.001154
## iter   5 value 5.001152
## iter   5 value 5.001152
```

```
## iter   5 value 5.001152
## final  value 5.001152
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE t.value p.value
## ma1   -0.4266 0.0756 -5.6471       0
## sar1  -0.5822 0.0740 -7.8712       0
## sar2  -0.4028 0.0763 -5.2760       0
##
## sigma^2 estimated as 21221.84 on 167 degrees of freedom
##
## AIC = 12.88724  AICc = 12.88809  BIC = 12.96102
##
```




```r
mod4$ttable # all terms are significant
```

```
##      Estimate     SE t.value p.value
## ma1   -0.4266 0.0756 -5.6471       0
## sar1  -0.5822 0.0740 -7.8712       0
## sar2  -0.4028 0.0763 -5.2760       0
```

Comments:

- Possible interpretations for the PACF and ACF:
  - p, q: ACF cuts off after 1 lags, PACF tails off.
  - p, q: ACF tails off, PACF cuts off after 2 lags (though the 2nd lag is not significant).

  - P, Q: ACF cuts off after 1 seasonal lag, PACF tails off.
  - P, Q: ACF tails off, PACF cuts off after 2 lags.
- All these models can be used.
- One typical residual analysis figure can be used here to illustrate the common advantages of the models.

Model selection

```
mod1$ICs
```

```
##      AIC     AICc      BIC
## 12.84862 12.84947 12.92241
```

```
mod2$ICs
```

```
##      AIC     AICc      BIC
## 12.84098 12.84140 12.89631
```

```
mod3$ICs
```

```
##      AIC     AICc      BIC
## 12.88304 12.88446 12.97527
```

```
mod4$ICs
```

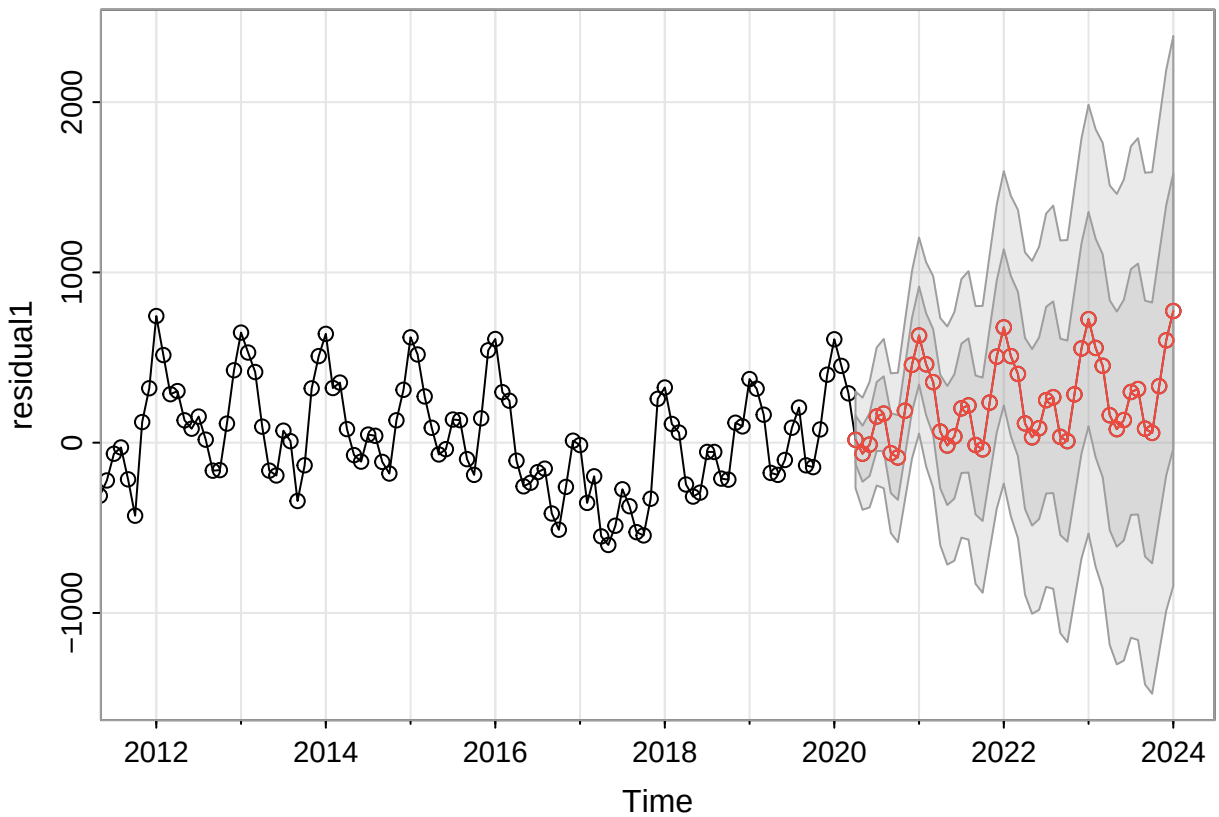
```
##      AIC     AICc      BIC
## 12.88724 12.88809 12.96102
```

Comments:

- Based on BIC, mod2 is the best.
- Based on AIC, mod2 is the best.
- mod2 is the best.

Forecasting using Regression 1 + mod2

```
# Forecast the length of test_data
n_forecasts <- nrow(test_data)
forecast_resid_1 <- sarima.for(residual1, n.ahead = n_forecasts,
                               p = 0, d = 1, q = 1, P = 0, D = 1, Q = 1, S = 12,
                               no.constant = TRUE)$pred
```

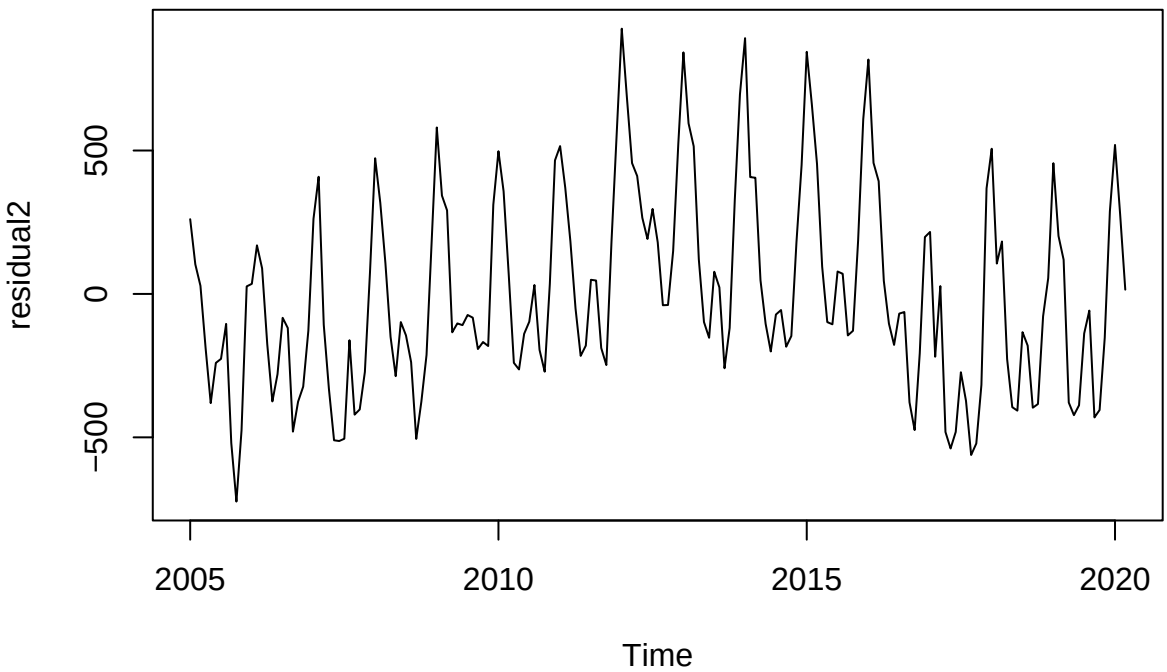



```
####### Note that the plot generated here is the forecast for RESIDUAL #######

regression_forecasts_1 <- predict(lm_model_1, newdata = test_data)

# Combine both forecasts
final_forecast_1 <- regression_forecasts_1 + forecast_resid_1

mse <- mean((test_data$gas - final_forecast_1)^2)
print(paste("The Mean Squared Error (MSE) of the forecasts is:", mse))
```

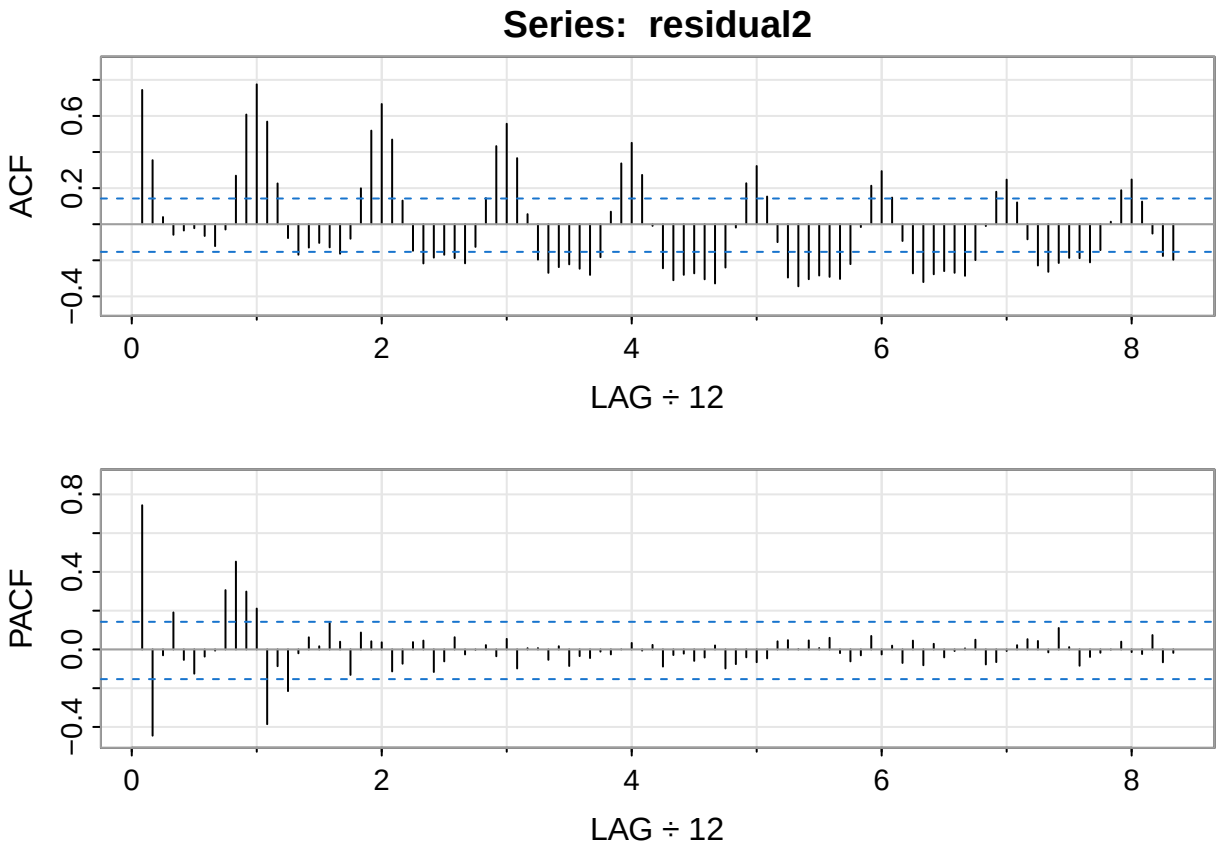
```
## [1] "The Mean Squared Error (MSE) of the forecasts is: 415001.476670315"
```

Transformation for residual2

```
# Plotting & ACF & PACF
ts.plot(residual2)
```




```r
acf2(residual2, max.lag = 100) # clear seasonality, with s = 12, clear non-stationarity
```




```
##      [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10] [,11] [,12]
## ACF  0.74  0.35  0.04 -0.06 -0.03 -0.02 -0.07 -0.12 -0.03  0.27  0.61  0.78
## PACF 0.74 -0.44 -0.03  0.19 -0.05 -0.13 -0.04 -0.01  0.31  0.45  0.30  0.21
##      [,13] [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24]
```

```
## ACF   0.57  0.23 -0.08 -0.17 -0.13 -0.10 -0.13 -0.16 -0.08  0.20  0.52  0.67
## PACF -0.39 -0.09 -0.22 -0.02  0.06  0.02  0.14  0.04 -0.13  0.09  0.04  0.04
##      [,25] [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36]
## ACF   0.47  0.13 -0.15 -0.22 -0.19 -0.17 -0.19 -0.22 -0.13  0.14  0.43  0.56
## PACF -0.11 -0.07  0.04  0.05 -0.12 -0.06  0.06 -0.03  0.00  0.02 -0.04  0.05
##      [,37] [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48]
## ACF   0.37  0.06 -0.20 -0.27 -0.24 -0.22 -0.25 -0.28 -0.18  0.07  0.34  0.45
## PACF -0.10  0.01  0.01 -0.05  0.02 -0.09 -0.03 -0.04 -0.01 -0.03  0.00  0.03
##      [,49] [,50] [,51] [,52] [,53] [,54] [,55] [,56] [,57] [,58] [,59] [,60]
## ACF   0.27 -0.01 -0.24 -0.31 -0.28 -0.27 -0.31 -0.33 -0.24 -0.02  0.23  0.32
## PACF -0.01  0.02 -0.09 -0.03 -0.02 -0.06 -0.04  0.02 -0.10 -0.08 -0.04 -0.07
##      [,61] [,62] [,63] [,64] [,65] [,66] [,67] [,68] [,69] [,70] [,71] [,72]
## ACF   0.15 -0.10 -0.30 -0.34 -0.30 -0.28 -0.29 -0.30 -0.22 -0.02  0.21  0.29
## PACF -0.05  0.04  0.05  0.00  0.05  0.01  0.06 -0.02 -0.06 -0.03  0.07 -0.03
##      [,73] [,74] [,75] [,76] [,77] [,78] [,79] [,80] [,81] [,82] [,83] [,84]
## ACF   0.15 -0.09 -0.27 -0.32 -0.28 -0.26 -0.27 -0.29 -0.20 -0.01  0.18  0.25
## PACF  0.02 -0.07  0.05 -0.08  0.03 -0.04 -0.01  0.01  0.05 -0.08 -0.07 -0.01
##      [,85] [,86] [,87] [,88] [,89] [,90] [,91] [,92] [,93] [,94] [,95] [,96]
## ACF   0.12 -0.08 -0.23 -0.26 -0.21 -0.19 -0.19 -0.21 -0.14  0.01  0.19  0.25
## PACF  0.02  0.05  0.04 -0.02  0.11  0.01 -0.08 -0.04 -0.02  0.00  0.04 -0.01
##      [,97] [,98] [,99] [,100]
## ACF   0.13 -0.05 -0.18  -0.20
## PACF -0.02  0.07 -0.07  -0.02
```

```r
## ACF is decaying very slowly!! need differencing

# adf test: stationary, but discard!
adf.test(residual2)
```

```
## Warning in adf.test(residual2): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  residual2
## Dickey-Fuller = -5.173, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

```r
# kpss test: not trend stationary & nearly not stationary
kpss.test(residual2, null = "Level") # not stationary on the significance level of 0.1
```

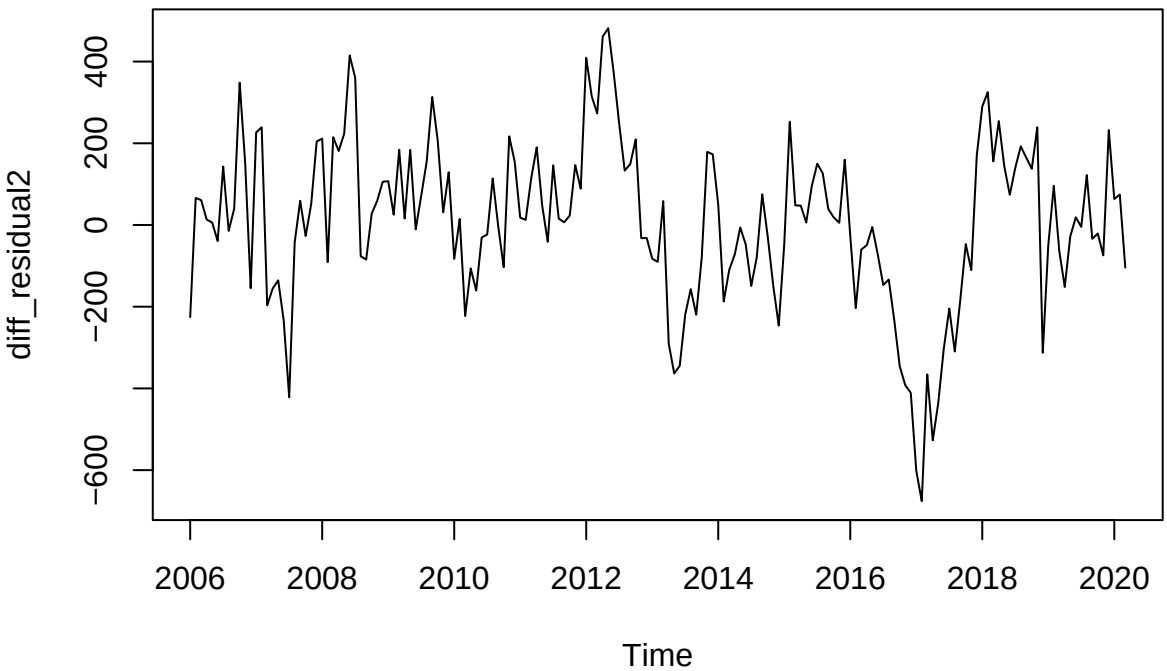
```
##
##  KPSS Test for Level Stationarity
##
## data:  residual2
## KPSS Level = 0.37445, Truncation lag parameter = 4, p-value = 0.08817
```

```r
kpss.test(residual2, null = "Trend") # not trend stationary
```

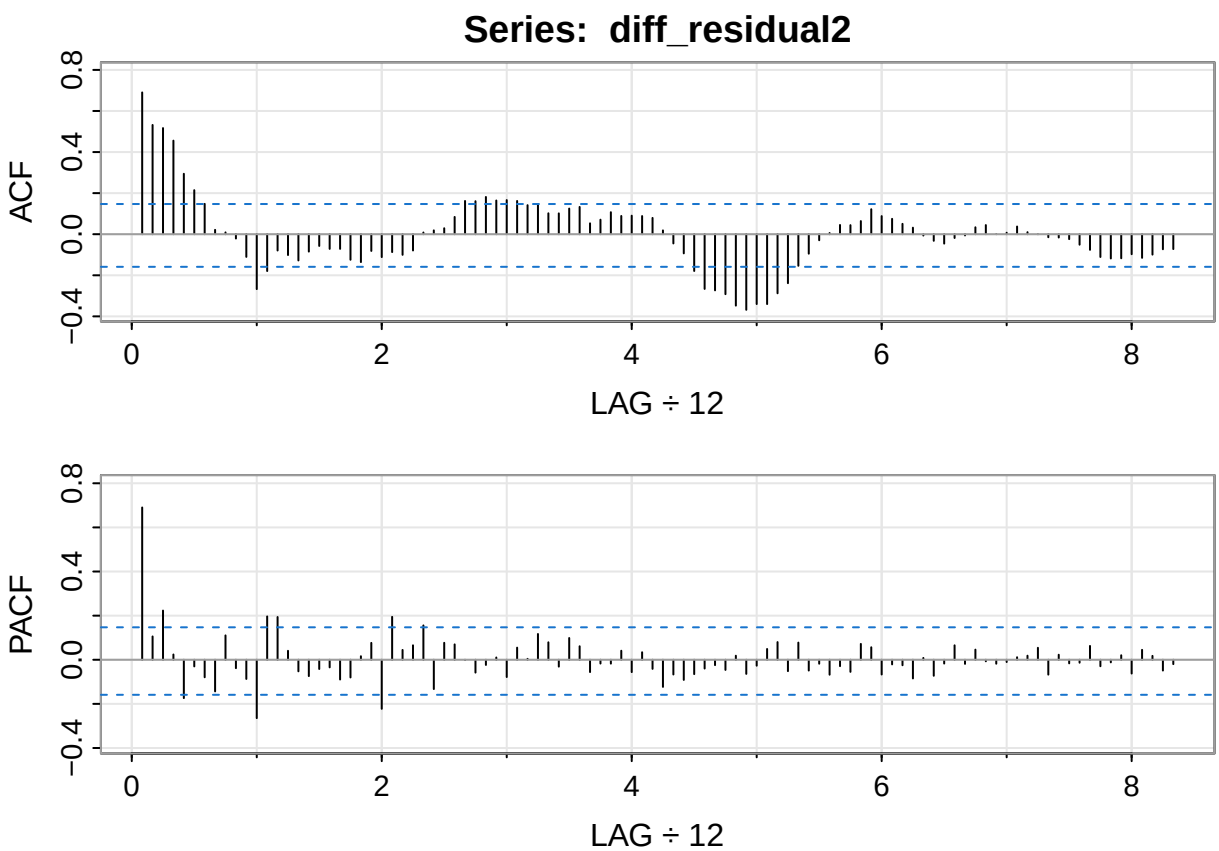
```
## Warning in kpss.test(residual2, null = "Trend"): p-value smaller than printed
## p-value
```

```
##
##  KPSS Test for Trend Stationarity
##
## data:  residual2
## KPSS Trend = 0.32112, Truncation lag parameter = 4, p-value = 0.01
```

```r
# Transformation: diff and seasonal diff
diff_residual2 = diff(residual2, 12) # seasonal differencing
ts.plot(diff_residual2)
```

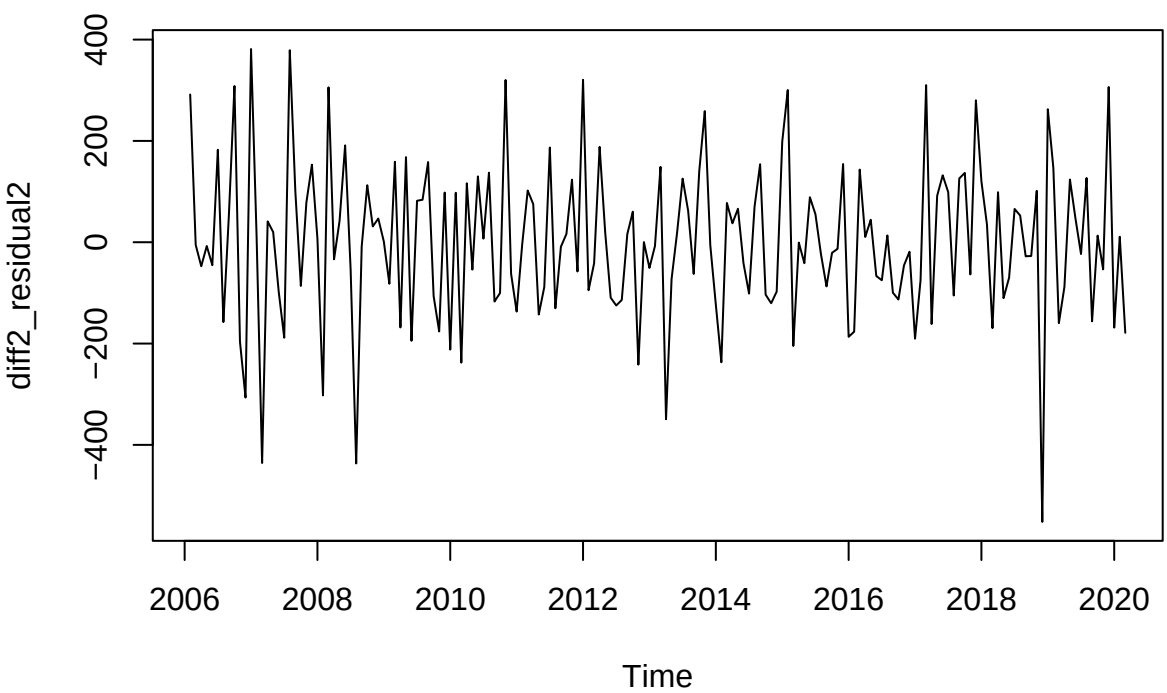



```r
acf2(diff_residual2, max.lag = 100) # ACF still decaying very slowly, another differencing
```

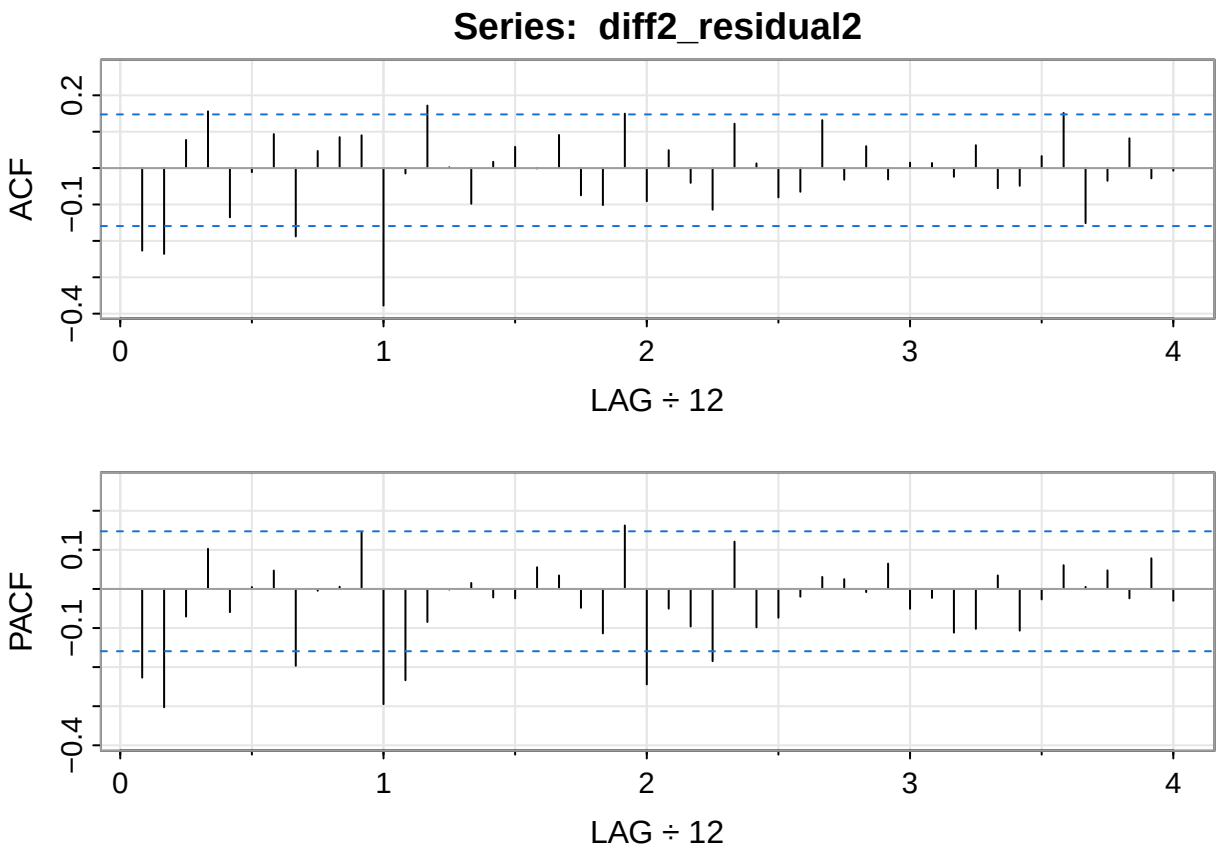


```
##      [,1] [,2] [,3] [,4]  [,5]  [,6]  [,7]  [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  0.69 0.53 0.52 0.46  0.29  0.21  0.15  0.02 0.01 -0.02 -0.11 -0.27 -0.18
## PACF 0.69 0.11 0.22 0.02 -0.17 -0.03 -0.08 -0.14 0.11 -0.04 -0.09 -0.26  0.20
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF  -0.08 -0.10 -0.13 -0.08 -0.06 -0.07 -0.07 -0.12 -0.14 -0.08 -0.11 -0.09
## PACF  0.19  0.04 -0.05 -0.07 -0.04 -0.03 -0.09 -0.08  0.02  0.08 -0.22  0.19
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF  -0.10 -0.08  0.01  0.02  0.03  0.08  0.16  0.16  0.18  0.16  0.17  0.16
## PACF  0.04  0.07  0.16 -0.13  0.08  0.07  0.00 -0.06 -0.02  0.01 -0.08  0.06
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48] [,49]
## ACF   0.14  0.14  0.10  0.10  0.12  0.13  0.05  0.07  0.11  0.09  0.09  0.09
## PACF  0.00  0.12  0.08 -0.03  0.10  0.06 -0.06 -0.02 -0.02  0.04 -0.06  0.03
##      [,50] [,51] [,52] [,53] [,54] [,55] [,56] [,57] [,58] [,59] [,60] [,61]
## ACF   0.08  0.02 -0.04 -0.09 -0.18 -0.27 -0.27 -0.29 -0.35 -0.37 -0.34 -0.34
## PACF -0.04 -0.12 -0.07 -0.09 -0.07 -0.04 -0.02 -0.05  0.02 -0.06 -0.03  0.05
##      [,62] [,63] [,64] [,65] [,66] [,67] [,68] [,69] [,70] [,71] [,72] [,73]
## ACF  -0.29 -0.24 -0.15 -0.09 -0.03  0.01  0.05  0.04  0.06  0.12  0.09  0.07
## PACF  0.08 -0.05  0.08 -0.05 -0.02 -0.07 -0.03 -0.05  0.07  0.06 -0.07 -0.02
##      [,74] [,75] [,76] [,77] [,78] [,79] [,80] [,81] [,82] [,83] [,84] [,85]
## ACF   0.05  0.03 -0.01 -0.03 -0.05 -0.02 -0.01  0.03  0.04  0.00  0.01  0.04
## PACF -0.02 -0.08  0.01 -0.07 -0.02  0.07 -0.02  0.05 -0.01 -0.02 -0.01  0.01
##      [,86] [,87] [,88] [,89] [,90] [,91] [,92] [,93] [,94] [,95] [,96] [,97]
## ACF   0.01  0.00 -0.01 -0.02 -0.02 -0.05 -0.08 -0.11 -0.12 -0.12 -0.10 -0.12
## PACF  0.02  0.05 -0.07  0.02 -0.02 -0.01  0.06 -0.03 -0.01  0.02 -0.06  0.05
##      [,98] [,99] [,100]
## ACF  -0.10 -0.07  -0.07
## PACF  0.02 -0.05  -0.02
```

```r
diff2_residual2 = diff(diff(residual2, 12))
ts.plot(diff2_residual2)
```



```r
acf2(diff2_residual2)
```



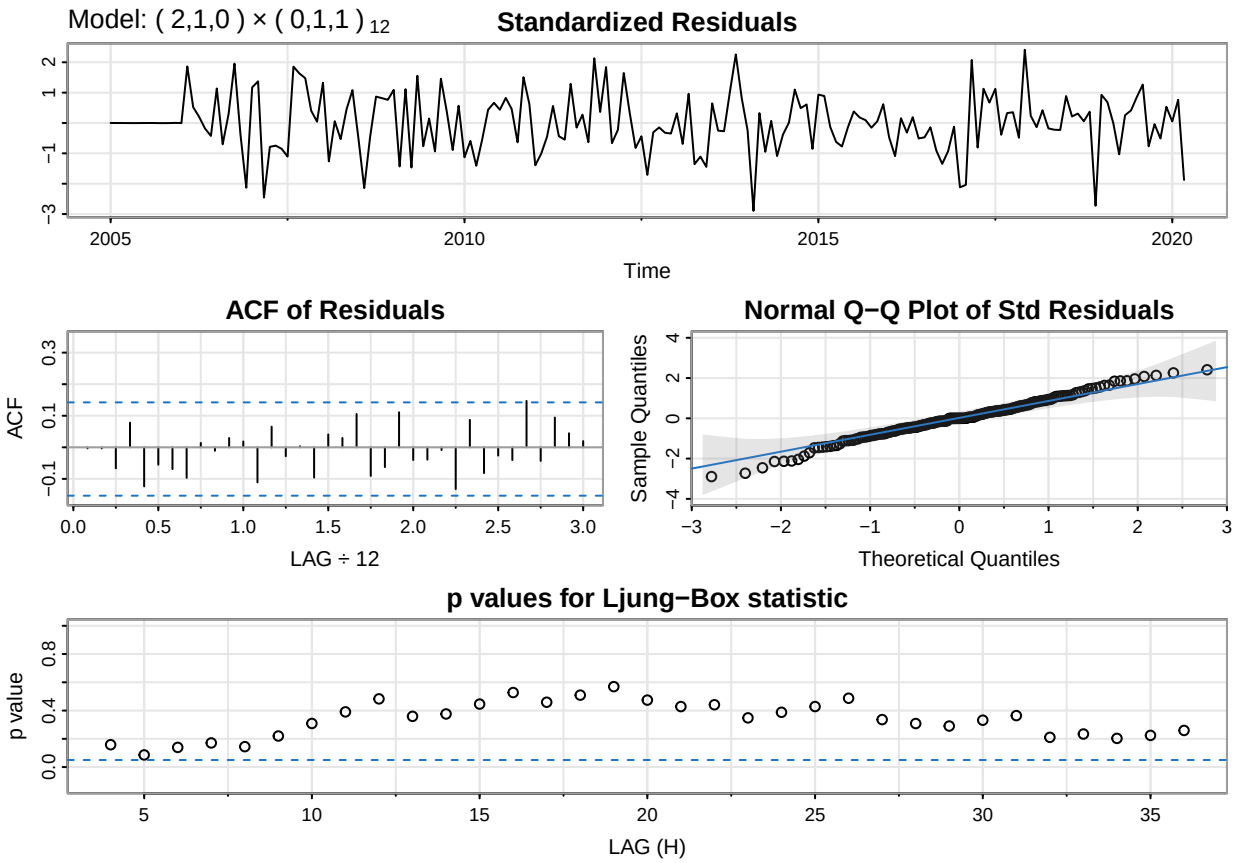

```
##       [,1]  [,2]  [,3] [,4]  [,5]  [,6] [,7]  [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  -0.23 -0.24  0.08 0.16 -0.14 -0.01 0.09 -0.19 0.05  0.09  0.09 -0.38 -0.01
## PACF -0.23 -0.30 -0.07 0.10 -0.06  0.00 0.05 -0.20 0.00  0.01  0.15 -0.29 -0.23
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF   0.17     0 -0.10  0.02  0.06  0.00  0.09 -0.07 -0.10  0.15 -0.09  0.05
## PACF -0.08     0  0.02 -0.02 -0.02  0.06  0.03 -0.05 -0.11  0.16 -0.24 -0.05
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF  -0.04 -0.11  0.12  0.01 -0.08 -0.07  0.13 -0.03  0.06 -0.03  0.01  0.01
## PACF -0.10 -0.18  0.12 -0.10 -0.07 -0.02  0.03  0.03 -0.01  0.06 -0.05 -0.02
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48]
## ACF  -0.02  0.06 -0.06 -0.05  0.03  0.15 -0.15 -0.03  0.08 -0.03 -0.01
## PACF -0.11 -0.10  0.03 -0.11 -0.03  0.06  0.01  0.05 -0.02  0.08 -0.03
```

Modeling for autocorrelation in residual2

```r
## First, try the interpretability of the acf pacf plots
mod1_2 = sarima(residual2, p=2, d=1, q=0, P=0, D=1, Q=1, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.041669
## iter   2 value 4.849533
## iter   3 value 4.828850
## iter   4 value 4.824074
## iter   5 value 4.813715
## iter   6 value 4.812141
```

```
## iter   7 value 4.811956
## iter   8 value 4.811941
## iter   9 value 4.811941
## iter   9 value 4.811940
## iter   9 value 4.811940
## final  value 4.811940
## converged
## initial  value 4.799981
## iter   2 value 4.794064
## iter   3 value 4.793676
## iter   4 value 4.793659
## iter   5 value 4.793658
## iter   5 value 4.793658
## iter   5 value 4.793658
## final  value 4.793658
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE  t.value p.value
## ar1   -0.3636 0.0746  -4.8722   0e+00
## ar2   -0.2703 0.0757  -3.5684   5e-04
## sma1  -0.7098 0.0642 -11.0617   0e+00
##
## sigma^2 estimated as 13855.59 on 167 degrees of freedom
##
## AIC = 12.47225  AICc = 12.4731  BIC = 12.54603
##
```



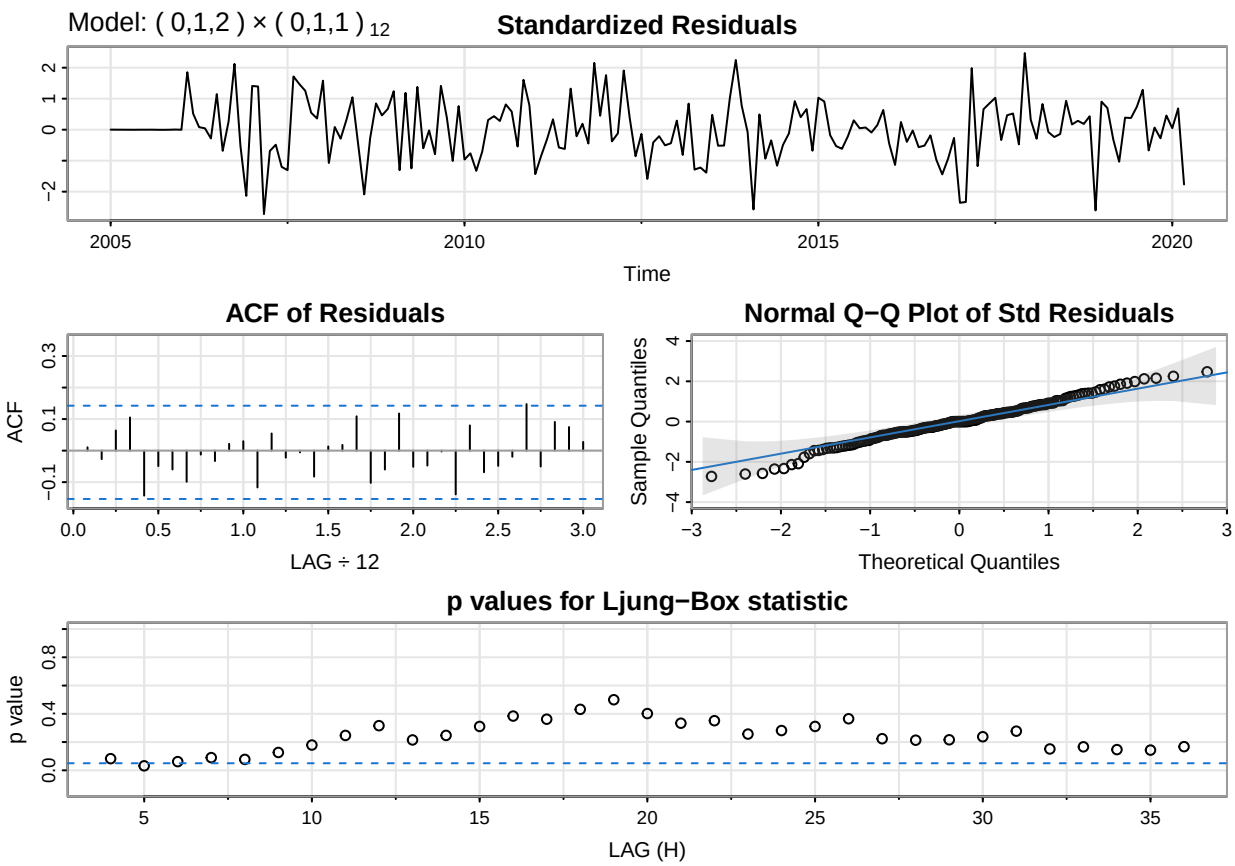

```r
mod1_2$ttable # all terms are significant
```

```
##      Estimate     SE  t.value p.value
## ar1   -0.3636 0.0746  -4.8722   0e+00
## ar2   -0.2703 0.0757  -3.5684   5e-04
## sma1  -0.7098 0.0642 -11.0617   0e+00
```

```r
mod2_2 = sarima(residual2, p=0, d=1, q=2, P=0, D=1, Q=1, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.046214
## iter   2 value 4.866508
## iter   3 value 4.843878
## iter   4 value 4.831682
## iter   5 value 4.825972
## iter   6 value 4.821652
## iter   7 value 4.821166
## iter   8 value 4.821131
## iter   9 value 4.821130
## iter  10 value 4.821130
## iter  10 value 4.821130
## iter  10 value 4.821130
## final  value 4.821130
## converged
## initial  value 4.800037
## iter   2 value 4.796413
## iter   3 value 4.795979
## iter   4 value 4.795946
```

```
## iter   5 value 4.795945
## iter   5 value 4.795945
## iter   5 value 4.795945
## final  value 4.795945
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE  t.value p.value
## ma1   -0.3773 0.0788  -4.7869  0.0000
## ma2   -0.1312 0.0860  -1.5258  0.1289
## sma1  -0.7352 0.0630 -11.6674  0.0000
##
## sigma^2 estimated as 13843.23 on 167 degrees of freedom
##
## AIC = 12.47683  AICc = 12.47768  BIC = 12.55061
##
```



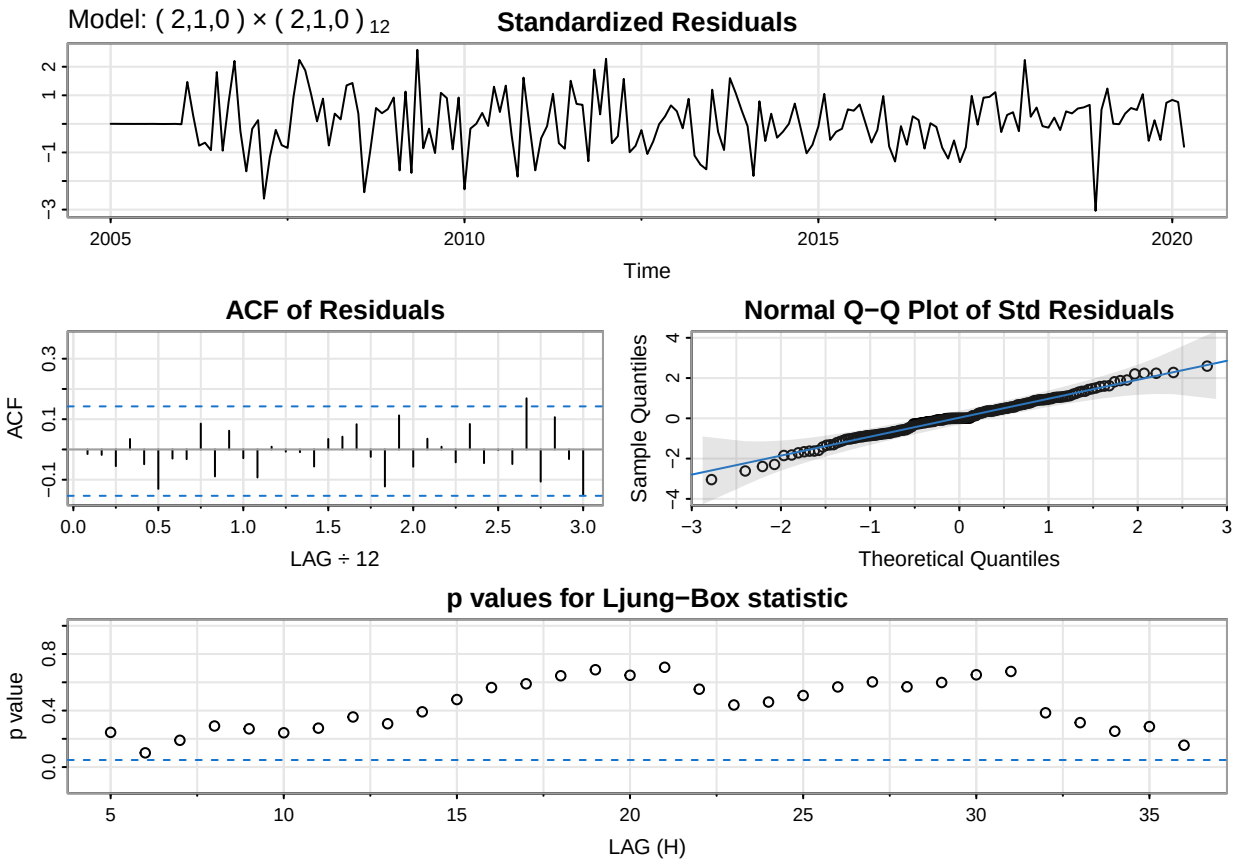

```r
mod2_2$ttable # ma2 is not significant
```

```
##      Estimate     SE  t.value p.value
## ma1   -0.3773 0.0788  -4.7869  0.0000
## ma2   -0.1312 0.0860  -1.5258  0.1289
## sma1  -0.7352 0.0630 -11.6674  0.0000
```

```r
mod3_2 = sarima(residual1, p=2, d=1, q=0, P=2, D=1, Q=0, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.166766
```

```
## iter   2 value 4.971743
## iter   3 value 4.930326
## iter   4 value 4.924673
## iter   5 value 4.924659
## iter   6 value 4.924575
## iter   7 value 4.924574
## iter   7 value 4.924574
## iter   7 value 4.924574
## final  value 4.924574
## converged
## initial  value 4.996773
## iter   2 value 4.993647
## iter   3 value 4.993204
## iter   4 value 4.993180
## iter   5 value 4.993169
## iter   5 value 4.993169
## iter   5 value 4.993169
## final  value 4.993169
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE t.value p.value
## ar1   -0.3595 0.0748 -4.8083   0e+00
## ar2   -0.2579 0.0752 -3.4312   8e-04
## sar1  -0.5885 0.0730 -8.0562   0e+00
## sar2  -0.4226 0.0750 -5.6311   0e+00
##
## sigma^2 estimated as 20826.94 on 166 degrees of freedom
##
## AIC = 12.88304  AICc = 12.88446  BIC = 12.97527
##
```



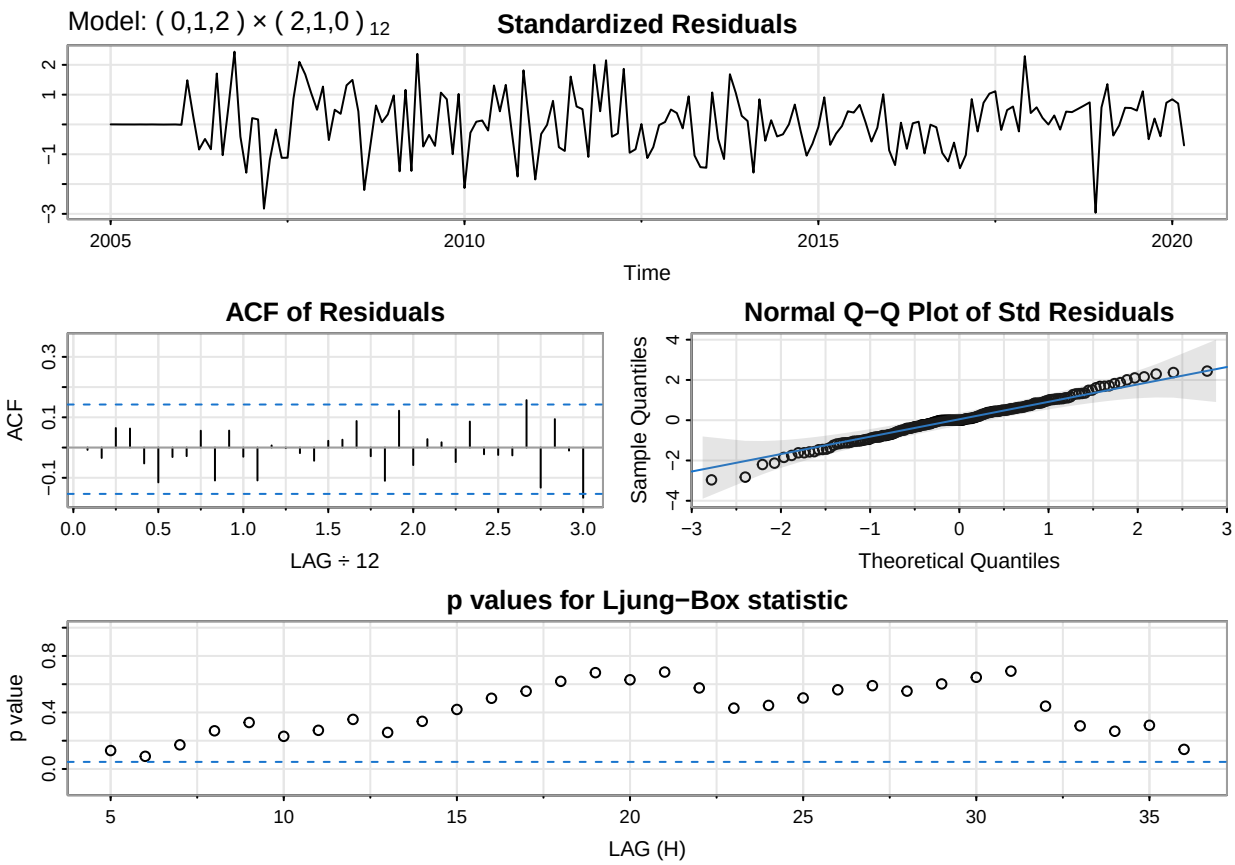

```r
mod3_2$ttable # all terms are significant
```

```
##      Estimate     SE t.value p.value
## ar1   -0.3595 0.0748 -4.8083   0e+00
## ar2   -0.2579 0.0752 -3.4312   8e-04
## sar1  -0.5885 0.0730 -8.0562   0e+00
## sar2  -0.4226 0.0750 -5.6311   0e+00
```

```r
mod4_2 = sarima(residual1, p=0, d=1, q=2, P=2, D=1, Q=0, S=12, no.constant = TRUE) # Ljung box is good
```

```
## initial  value 5.179073
## iter   2 value 4.955231
## iter   3 value 4.929144
## iter   4 value 4.917318
## iter   5 value 4.913619
## iter   6 value 4.912429
## iter   7 value 4.912156
## iter   8 value 4.912105
## iter   9 value 4.912104
## iter   9 value 4.912104
## iter   9 value 4.912104
## final  value 4.912104
## converged
## initial  value 4.999552
## iter   2 value 4.997320
## iter   3 value 4.995227
## iter   4 value 4.995065
```

```
## iter   5 value 4.995060
## iter   6 value 4.995059
## iter   7 value 4.995059
## iter   7 value 4.995059
## iter   7 value 4.995059
## final  value 4.995059
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate     SE t.value p.value
## ma1   -0.3667 0.0792 -4.6321  0.0000
## ma2   -0.1247 0.0849 -1.4693  0.1436
## sar1  -0.5781 0.0732 -7.8948  0.0000
## sar2  -0.4170 0.0757 -5.5043  0.0000
##
## sigma^2 estimated as 20930.81 on 166 degrees of freedom
##
## AIC = 12.88682  AICc = 12.88824  BIC = 12.97905
##
```




```r
mod4_2$ttable # ma2 is not significant
```

```
##      Estimate     SE t.value p.value
## ma1   -0.3667 0.0792 -4.6321  0.0000
## ma2   -0.1247 0.0849 -1.4693  0.1436
## sar1  -0.5781 0.0732 -7.8948  0.0000
```

```
## sar2  -0.4170 0.0757 -5.5043  0.0000
```

Comments:

- Possible interpretations for the PACF and ACF:
  - p, q: ACF cuts off after 2 lags, PACF tails off.
  - p, q: ACF tails off, PACF cuts off after 2 lags (though the 2nd lag is not significant).
  - P, Q: ACF cuts off after 1 seasonal lag, PACF tails off.
  - P, Q: ACF tails off, PACF cuts off after 2 lags.
- All these models can be used to compare AIC / BIC.
- One typical residual analysis figure can be used here to illustrate the disadvantage of some models. (e.g., some term is not significant)

Model selection

```r
mod1_2$ICs
```

```
##      AIC     AICc      BIC
## 12.47225 12.47310 12.54603
```

```r
mod2_2$ICs
```

```
##      AIC     AICc      BIC
## 12.47683 12.47768 12.55061
```

```r
mod3_2$ICs
```

```
##      AIC     AICc      BIC
## 12.88304 12.88446 12.97527
```

```r
mod4_2$ICs
```

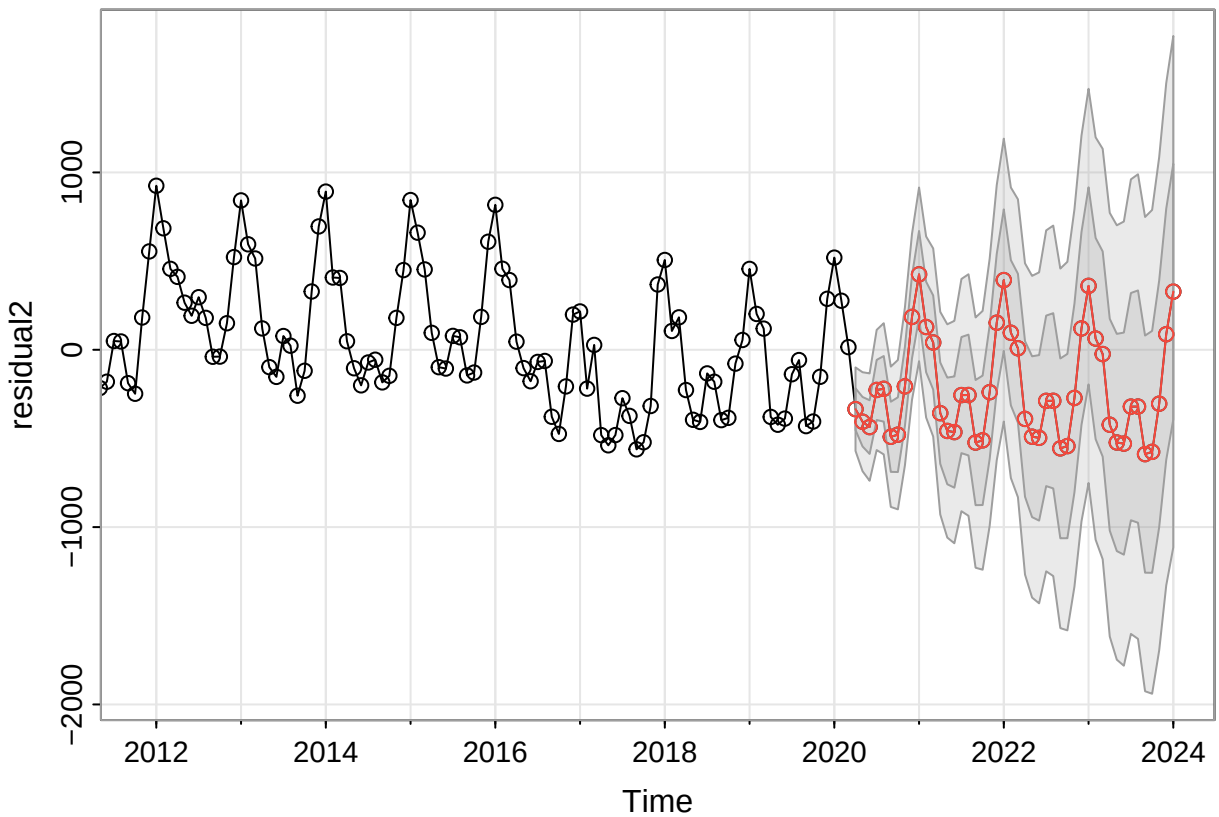
```
##      AIC     AICc      BIC
## 12.88682 12.88824 12.97905
```

Comments:

- Based on BIC, mod1 is the best.
- Based on AIC, mod1 is the best.
- mod1 is the best.

Forecasting using Regression 2 + mod1_2

```r
# Forecast the length of test_data
n_forecasts <- nrow(test_data)
forecast_resid_2 <- sarima.for(residual2, n.ahead = n_forecasts,
                               p = 2, d = 1, q = 0, P = 0, D = 1, Q = 1, S = 12,
                               no.constant = TRUE)$pred
```




```
####### Note that the plot generated here is the forecast for RESIDUALS #######

regression_forecasts_2 <- predict(lm_model_2, newdata = test_data)

# Combine both forecasts
final_forecast_2 <- regression_forecasts_2 + forecast_resid_2

mse <- mean((test_data$gas - final_forecast_2)^2)
print(paste("The Mean Squared Error (MSE) of the forecasts is:", mse))
```

```
## [1] "The Mean Squared Error (MSE) of the forecasts is: 496704.480839348"
```

Final Conclusions

- Based on the forecasting MSE, Regression 1 + mod 2 is the best model overall.
- When writing report, the AIC and BIC of the two regression models can also be taken into account.

```
list(
  AIC_1 = AIC(lm_model_1),
  BIC_1 = BIC(lm_model_1)
)
```

```
## $AIC_1
## [1] 2618.499
##
## $BIC_1
## [1] 2637.756
```

```r
list(
  AIC_2 = AIC(lm_model_2),
  BIC_2 = BIC(lm_model_2)
)
```

```
## $AIC_2
## [1] 2663.479
##
## $BIC_2
## [1] 2682.736
```

- Based on AIC and BIC of the pure regression models, we still choose regression 1.

Forecasting for the next 5 periods using our best model:

```r
head(final_forecast_1, 5)
```

```
##        1        2        3        4        5
## 2374.753 2167.899 2128.990 2360.613 2505.254
```